Chapter 13

Kalman Filters

13.1 Introduction

We now discuss an important generalization of the Wiener filter. The significance of the extension is in its ability to accommodate vector signals and noises which additionally may be nonstationary. This is in contrast to the Wiener filter, which is restricted to stationary scalar signals and noises. This generalization is termed the *Kalman filter*. It may be thought of as a sequential MMSE estimator of a signal embedded in noise, where the signal is characterized by a dynamical or state model. It generalizes the sequential MMSE estimator in Section 12.6, to allow the unknown parameters to evolve in time according to a dynamical model. If the signal and noise are jointly Gaussian, then the Kalman filter is an optimal MMSE estimator, and if not, it is the optimal LMMSE estimator.

13.2 Summary

The scalar Gauss-Markov signal model is given in recursive form by (13.1) and explicitly by (13.2). Its mean, covariance, and variance are given by (13.4), (13.5), and (13.6), respectively. Generalizing the model to a vector signal results in (13.12) with the statistical assumptions summarized just below. Also, the explicit representation is given in (13.13). The corresponding mean and covariances are (13.14), (13.15), and (13.16) or in recursive form by (13.17) and (13.18). A summary of the vector Gauss-Markov signal model is given in Theorem 13.1. The sequential MMSE estimator or Kalman filter for a scalar signal and scalar observations is given by (13.38)–(13.42). If the filter attains steady-state, then it becomes the infinite length Wiener filter as described in Section 13.5. Two generalizations of the Kalman filter are to a vector signal, whose equations are (13.50)–(13.54), and to a vector signal and vector observations, whose assumptions and implementation are described in Theorem 13.2. When the signal model and/or observation model is nonlinear, the preceding Kalman filter cannot be applied directly. However, using a linearization approach, one obtains the suboptimal extended Kalman filter, whose equations are (13.67)–(13.71).

13.3 Dynamical Signal Models

We begin our discussion of signal modeling by recalling our usual example of a DC level in WGN or

$$x[n] = A + w[n]$$

where A is the parameter to be estimated and $w[n]$ is WGN with variance σ^2. The signal model A might represent the voltage output of a DC power supply, and the noise $w[n]$ might model the error introduced by an inaccurate voltmeter as successive measurements are taken. Hence, $x[n]$ represents the noise corrupted observations of the power supply output. Now, even though the power supply should generate a constant voltage of A, in practice the true voltage will vary slightly as time progresses. This is due to the effects of temperature, component aging, etc., on the circuitry. Hence, a more accurate measurement model would be

$$x[n] = A[n] + w[n]$$

where $A[n]$ is the true voltage at time n. However, with this model our estimation problem becomes considerably more complicated since we will need to estimate $A[n]$ for $n = 0, 1, \ldots, N-1$ instead of just the single parameter A. To underscore the difficulty assume that we model the voltage $A[n]$ as a sequence of unknown deterministic parameters. Then, the MVU estimator of $A[n]$ is easily shown to be

$$\hat{A}[n] = x[n].$$

The estimates will be inaccurate due to a lack of averaging, and in fact the variability will be identical to that of the noise or $\text{var}(\hat{A}[n]) = \sigma^2$. This estimator is undesirable in that it allows estimates such as those shown in Figure 13.1. We can expect that if the power supply is set for $A = 10$ volts, then the true voltage will be near this and the variation over time will be slow (otherwise it's time to buy a new power supply!). An example of the true voltage is given in Figure 13.1 and is seen to vary about 10 volts. Successive samples of $A[n]$ will not be too different, leading us to conclude that they display a high degree of "correlation." This reasoning naturally leads us to consider $A[n]$ as a realization of a random process with a mean of 10 and some correlation between samples. The imposition of a correlation constraint will prevent the estimate of $A[n]$ from fluctuating too wildly in time. Thus, we will consider $A[n]$ to be a realization of a random process to be estimated, for which Bayesian approaches are appropriate. This type of modeling was used in Chapter 12 in our discussion of Wiener filtering. There the signal to be estimated was termed $s[n]$, and it was assumed to be zero mean. In keeping with this standard notation, we now adopt $s[n]$ as our notation as opposed to $\theta[n]$. Also, because of the zero mean assumption, $s[n]$ will represent the signal model for $A[n] - 10$. Once we specify the signal model for zero mean $s[n]$, it is easily modified to accommodate nonzero mean processes by adding $E(s[n])$ to it. We will always assume that the mean is *known*.

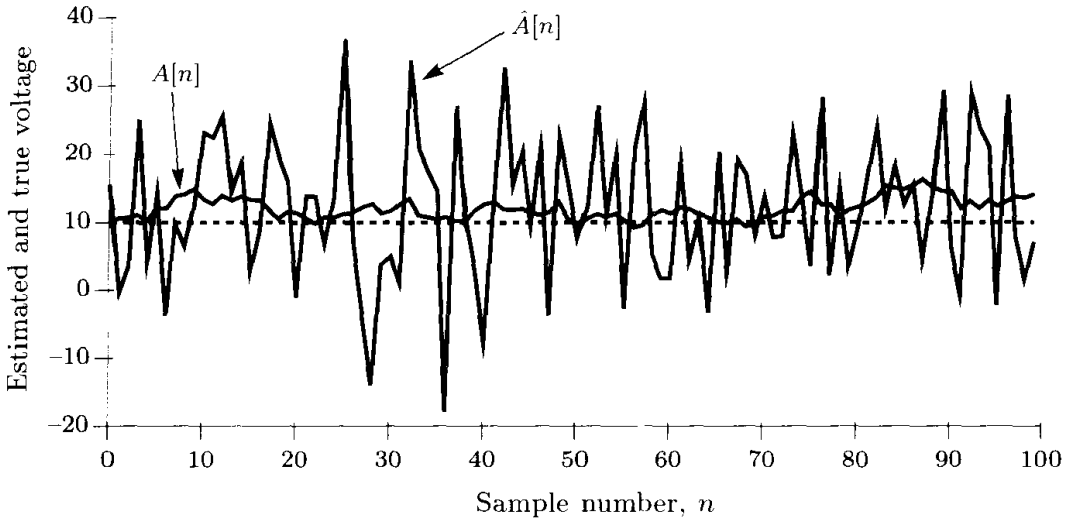


Figure 13.1 True voltage and MVU estimator

A simple model for $s[n]$ which allows us to specify the correlation between samples is the *first-order Gauss-Markov* process

$$s[n] = as[n-1] + u[n] \qquad n \geq 0 \tag{13.1}$$

where $u[n]$ is WGN with variance σ_u^2, $s[-1] \sim \mathcal{N}(\mu_s, \sigma_s^2)$, and $s[-1]$ is independent of $u[n]$ for all $n \geq 0$. (The reader should not confuse the Gauss-Markov *process* with the model considered in the Gauss-Markov *theorem* since they are different.) The noise $u[n]$ is termed the *driving or excitation noise* since $s[n]$ may be viewed as the output of a linear time invariant system driven by $u[n]$. In the control literature the system is referred to as the *plant*, and $u[n]$ is termed the *plant noise* [Jaswinski 1970]. The model of (13.1) is also called the *dynamical* or *state model*. The current output $s[n]$ depends only on the state of the system at the previous time, or $s[n-1]$, and the current input $u[n]$. The state of a system at time n_0 is generally defined to be the amount of information, which together with the input for $n \geq n_0$ determines the output for $n \geq n_0$ [Chen 1970]. Clearly, the state is $s[n-1]$, and it summarizes the effect of all past inputs to the system. We will henceforth refer to (13.1) as the Gauss-Markov model, where it is understood to be first order.

The signal model of (13.1) resembles an AR(1) process except that the signal starts at $n = 0$, and hence may not be WSS. We shall see shortly that as $n \to \infty$, so that the effect of the initial condition is negligible, the process is actually WSS and may be regarded as an AR(1) process with filter parameter $a[1] = -a$. A typical realization of $s[n]$ is shown in Figure 13.2 for $a = 0.98$, $\sigma_u^2 = 0.1$, $\mu_s = 5$, $\sigma_s^2 = 1$. Note that the mean starts off at about 5 but quickly decreases to zero. This behavior is just the transient

response of the system to the large initial sample $s[-1] \approx \mu_s = 5$. Also, the samples are heavily correlated. We may quantify these results by determining the mean and covariance of $s[n]$. Then, a complete statistical description will have been specified since $s[n]$ is a Gaussian process, as we will now show.

First, we express $s[n]$ as a function of the initial condition, and the inputs as

$$\begin{aligned} s[0] &= as[-1] + u[0] \\ s[1] &= as[0] + u[1] \\ &= a^2 s[-1] + au[0] + u[1] \\ &\text{etc.} \end{aligned}$$

In general, we have

$$s[n] = a^{n+1}s[-1] + \sum_{k=0}^{n} a^k u[n-k] \tag{13.2}$$

and we see that $s[n]$ is a linear function of the initial condition and the driving noise inputs from the initial time to the present time. Since these random variables are all independent and Gaussian, $s[n]$ is Gaussian, and it may further be shown that $s[n]$ is a Gaussian random process (see Problem 13.1). Now, the mean follows from (13.2) as

$$\begin{aligned} E(s[n]) &= a^{n+1}E(s[-1]) &(13.3)\\ &= a^{n+1}\mu_s. &(13.4) \end{aligned}$$

The covariance between samples $s[m]$ and $s[n]$ is, from (13.2) and (13.4),

$$\begin{aligned} c_s[m,n] &= E\left[(s[m] - E(s[m]))(s[n] - E(s[n]))\right] \\ &= E\left[\left(a^{m+1}(s[-1]-\mu_s) + \sum_{k=0}^{m} a^k u[m-k]\right)\right. \\ &\quad \left.\cdot\left(a^{n+1}(s[-1]-\mu_s) + \sum_{l=0}^{n} a^l u[n-l]\right)\right] \\ &= a^{m+n+2}\sigma_s^2 + \sum_{k=0}^{m}\sum_{l=0}^{n} a^{k+l}E(u[m-k]u[n-l]). \end{aligned}$$

But

$$E(u[m-k]u[n-l]) = \sigma_u^2\delta[l-(n-m+k)]$$

or

$$E(u[m-k]u[n-l]) = \begin{cases} \sigma_u^2 & l = n-m+k \\ 0 & \text{otherwise.} \end{cases}$$

Thus, for $m \geq n$

$$\begin{aligned} c_s[m,n] &= a^{m+n+2}\sigma_s^2 + \sigma_u^2 \sum_{k=m-n}^{m} a^{2k+n-m} \\ &= a^{m+n+2}\sigma_s^2 + \sigma_u^2 a^{m-n}\sum_{k=0}^{n} a^{2k} &(13.5) \end{aligned}$$

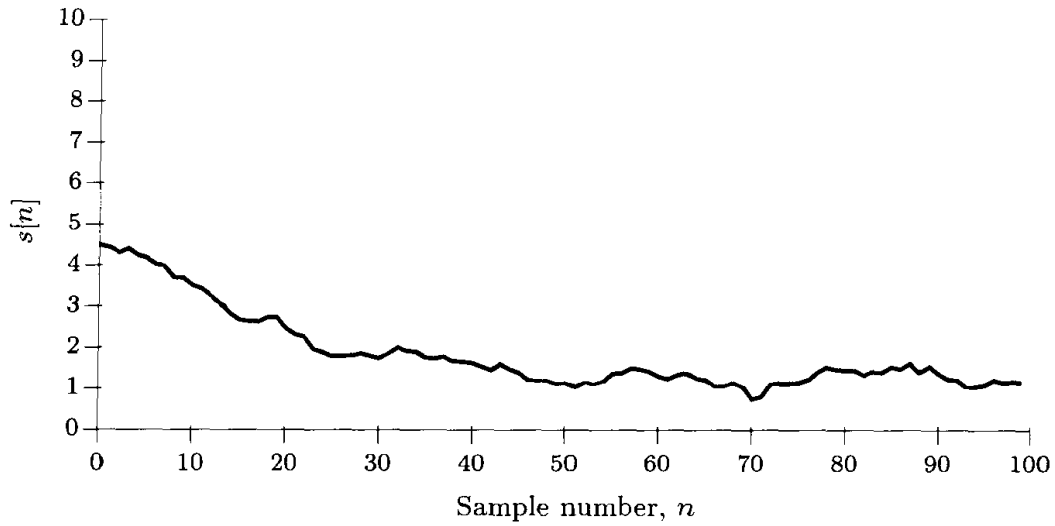


Figure 13.2 Typical realization of first order Gauss-Markov process

and of course $c_s[m,n] = c_s[n,m]$ for $m < n$. Note that the variance is

$$\begin{aligned} \text{var}(s[n]) &= c_s[n,n] \\ &= a^{2n+2}\sigma_s^2 + \sigma_u^2 \sum_{k=0}^{n} a^{2k}. \end{aligned} \tag{13.6}$$

Clearly, $s[n]$ is not WSS since the mean depends on n and the covariance depends on m and n, not the difference. However, as $n \to \infty$, we have from (13.4) and (13.5)

$$\begin{aligned} E(s[n]) &\to 0 \\ c_s[m,n] &\to \frac{\sigma_u^2 a^{m-n}}{1-a^2} \end{aligned}$$

since $|a| < 1$, which is necessary for stability of the process. Otherwise, the mean and variance would increase exponentially with n. Thus, as $n \to \infty$, the mean is zero and the covariance becomes an ACF with

$$r_{ss}[k] = c_s[m,n]|_{m-n=k} = \frac{\sigma_u^2}{1-a^2}a^k \qquad k \geq 0$$

which is recognized as the ACF of an AR(1) process. It is even possible that $s[n]$ will be WSS for $n \geq 0$ if the initial conditions are suitably chosen (see Problem 13.2). In any event it is seen from (13.5) that by properly choosing a the process can be made heavily correlated ($|a| \to 1$) or not correlated ($|a| \to 0$). This is also apparent from (13.1). As an illustration, for the process described earlier we plot the mean, variance, and steady-state covariance or ACF in Figure 13.3.

Because of the special form of the Gauss-Markov process, the mean and variance can also be expressed recursively. This is useful for conceptualization purposes as well as for extending the results to the vector Gauss-Markov process. The mean and variance are obtained directly from (13.1) as

$$E(s[n]) = aE(s[n-1]) + E(u[n])$$

or

$$E(s[n]) = aE(s[n-1]) \tag{13.7}$$

and

$$\begin{aligned} \text{var}(s[n]) &= E\left[(s[n] - E(s[n]))^2\right] \\ &= E\left[(as[n-1] + u[n] - aE(s[n-1]))^2\right] \\ &= a^2\text{var}(s[n-1]) + \sigma_u^2 \end{aligned} \tag{13.8}$$

where we have used $E(u[n]s[n-1]) = 0$. This follows from (13.2) because $s[n-1]$ depends only on $\{s[-1], u[0], \ldots, u[n-1]\}$ and they are independent of $u[n]$ by assumption. The equations (13.7) and (13.8) are termed the *mean and variance propagation equations.* A covariance propagation equation is explored in Problem 13.4. Note from (13.8) that the variance is decreased due to the effect of $as[n-1]$ but increased due to $u[n]$. In steady-state or as $n \to \infty$, these effects balance each other to yield the variance $\sigma_u^2/(1-a^2)$ (just let $\text{var}(s[n-1]) = \text{var}(s[n])$ in (13.8) and solve).

The ease with which we were able to obtain the mean and variance propagation equations resulted from the dependence of $s[n]$ on only the previous sample $s[n-1]$ and the input $u[n]$. Consider now a pth-order Gauss-Markov process expressed in a notation reminiscent of an AR(p) process or

$$s[n] = -\sum_{k=1}^{p} a[k]s[n-k] + u[n]. \tag{13.9}$$

Because $s[n]$ now depends on the p previous samples, the mean and variance propagation equations become more complicated. To extend our previous results we first note that the state of the system at time n is $\{s[n-1], s[n-2], \ldots, s[n-p]\}$ since the previous p samples together with $u[n]$ determine the output. We thus define the *state vector* as

$$\mathbf{s}[n-1] = \begin{bmatrix} s[n-p] \\ s[n-p+1] \\ \vdots \\ s[n-1] \end{bmatrix}. \tag{13.10}$$

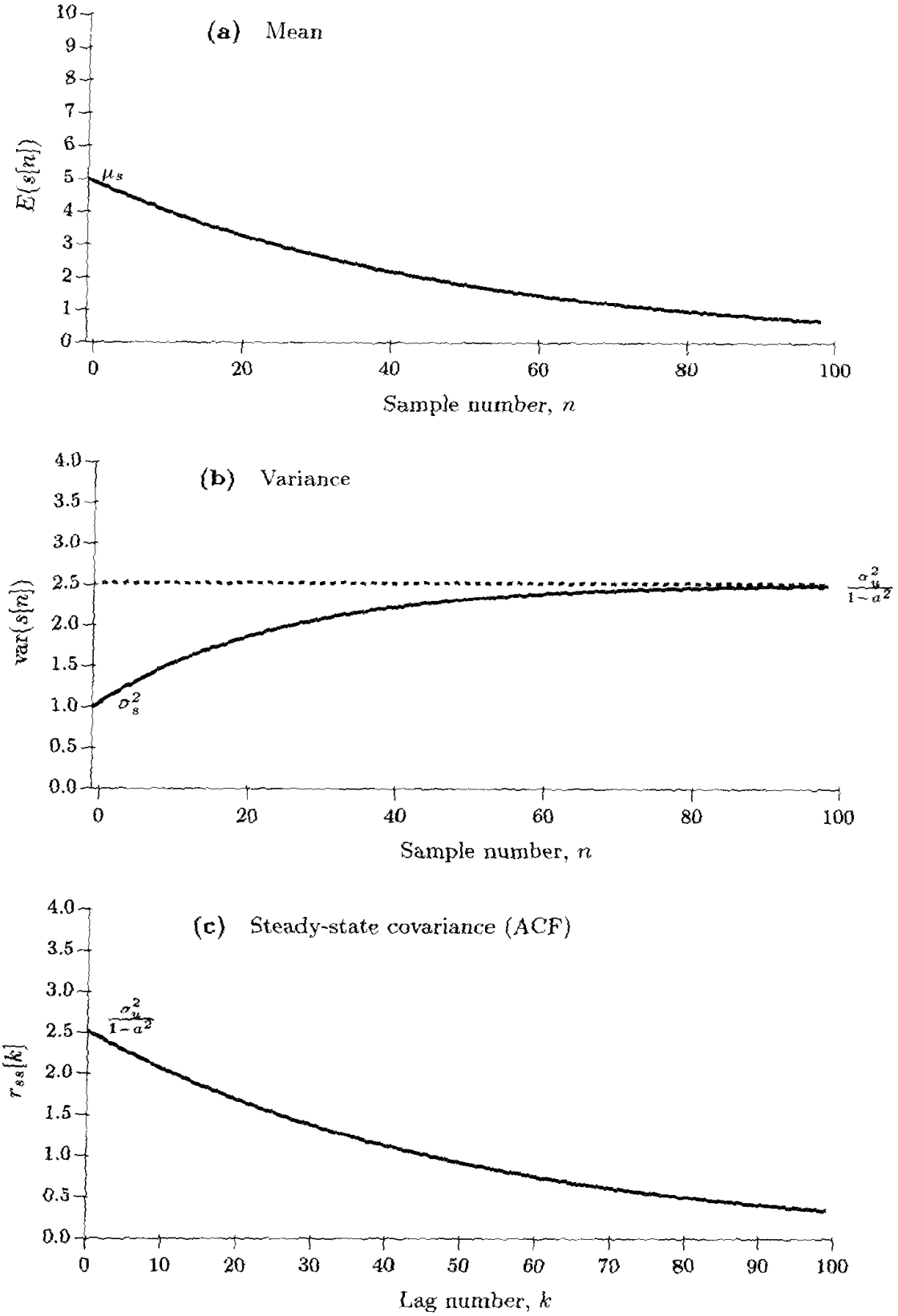


Figure 13.3 Statistics of first-order Gauss-Markov process

With this definition we can rewrite (13.9) in the form

$$\begin{bmatrix} s[n-p+1] \\ s[n-p+2] \\ \vdots \\ s[n-1] \\ s[n] \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a[p] & -a[p-1] & -a[p-2] & \dots & -a[1] \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} s[n-p] \\ s[n-p+1] \\ \vdots \\ s[n-2] \\ s[n-1] \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{B}} u[n]$$

where the additional $(p-1)$ equations are just identities. Hence, using the definition of the state vector, we have

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}u[n] \tag{13.11}$$

where $\mathbf{A}$ is a $p \times p$ nonsingular matrix (termed the *state transition matrix*) and $\mathbf{B}$ is a $p \times 1$ vector. Now we have the desired form (compare this to (13.1)) in which the vector signal $\mathbf{s}[n]$ is easily computed based on its value at the previous time instant, the state vector, and the input. This is termed the *vector* Gauss-Markov model. A final level of generality allows the input $u[n]$ to be an $r \times 1$ vector so that, as shown in Figure 13.4, we have a model for a vector signal as the output of a linear time invariant system ($\mathbf{A}$ and $\mathbf{B}$ are constant matrices) excited by a vector input. In Figure 13.4a $\mathcal{H}(z)$ is the $p \times r$ matrix system function. Summarizing, our general vector Gauss-Markov model takes the form

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] \qquad n \geq 0 \tag{13.12}$$

where $\mathbf{A}$, $\mathbf{B}$ are constant matrices with dimension $p \times p$ and $p \times r$, respectively, $\mathbf{s}[n]$ is the $p \times 1$ signal vector, and $\mathbf{u}[n]$ is the $r \times 1$ driving noise vector. We will from time to time refer to (13.12) as the *state model*. The statistical assumptions are that

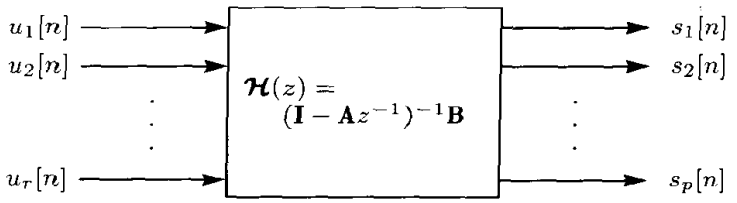
1. The input $\mathbf{u}[n]$ is a vector WGN sequence, i.e., $\mathbf{u}[n]$ is a sequence of uncorrelated jointly Gaussian vectors with $E(\mathbf{u}[n]) = \mathbf{0}$. As a result, we have that

$$E(\mathbf{u}[m]\mathbf{u}^T[n]) = \mathbf{0} \qquad m \neq n$$

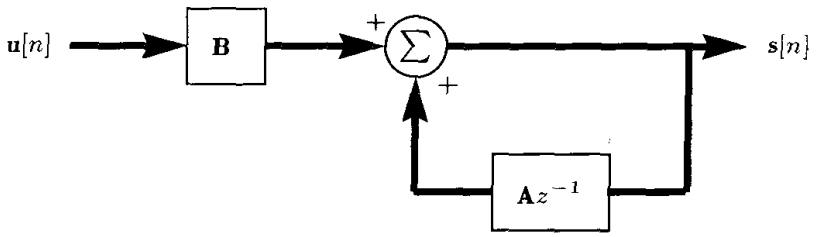
and the covariance of $\mathbf{u}[n]$ is

$$E(\mathbf{u}[n]\mathbf{u}^T[n]) = \mathbf{Q},$$

where $\mathbf{Q}$ is an $r \times r$ positive definite matrix. Note that the vector samples are independent due to the jointly Gaussian assumption.



(a) Multi-input—multi-output model



(b) Equivalent vector model

Figure 13.4 Vector Gauss-Markov signal system model

2. The initial state or $\mathbf{s}[-1]$ is a random vector with

$$\mathbf{s}[-1] \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$$

and is independent of $\mathbf{u}[n]$ for all $n \geq 0$.

We now illustrate with an example.

Example 13.1 - Two DC Power Supplies

Recalling the introductory example, consider now the model for the outputs of two DC power supplies that vary with time. If we assume that the outputs are independent (in a functional sense) of each other, then a reasonable model would be the scalar model of (13.1) for each output or

$$\begin{aligned} s_1[n] &= a_1 s_1[n-1] + u_1[n] \\ s_2[n] &= a_2 s_2[n-1] + u_2[n] \end{aligned}$$

where $s_1[-1] \sim \mathcal{N}(\mu_{s_1}, \sigma_{s_1}^2)$, $s_2[-1] \sim \mathcal{N}(\mu_{s_2}, \sigma_{s_2}^2)$, $u_1[n]$ is WGN with variance $\sigma_{u_1}^2$, $u_2[n]$ is WGN with variance $\sigma_{u_2}^2$, and all random variables are independent of each other. Considering $\mathbf{s}[n] = [s_1[n]\, s_2[n]]^T$ as the vector parameter to be estimated, we have the model

$$\underbrace{\begin{bmatrix} s_1[n] \\ s_2[n] \end{bmatrix}}_{\mathbf{s}[n]} = \underbrace{\begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} s_1[n-1] \\ s_2[n-1] \end{bmatrix}}_{\mathbf{s}[n-1]} + \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\mathbf{B}} \underbrace{\begin{bmatrix} u_1[n] \\ u_2[n] \end{bmatrix}}_{\mathbf{u}[n]}$$

where $\mathbf{u}[n]$ is vector WGN with zero mean and covariance

$$\begin{aligned} E(\mathbf{u}[m]\mathbf{u}^T[n]) &= \begin{bmatrix} E(u_1[m]u_1[n]) & E(u_1[m]u_2[n]) \\ E(u_2[m]u_1[n]) & E(u_2[m]u_2[n]) \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{u_1}^2 & 0 \\ 0 & \sigma_{u_2}^2 \end{bmatrix} \delta[m-n] \end{aligned}$$

so that

$$\mathbf{Q} = \begin{bmatrix} \sigma_{u_1}^2 & 0 \\ 0 & \sigma_{u_2}^2 \end{bmatrix}$$

and

$$\begin{aligned} \mathbf{s}[-1] &= \begin{bmatrix} s_1[-1] \\ s_2[-1] \end{bmatrix} \\ &\sim \mathcal{N}\left(\begin{bmatrix} \mu_{s_1} \\ \mu_{s_2} \end{bmatrix}, \begin{bmatrix} \sigma_{s_1}^2 & 0 \\ 0 & \sigma_{s_2}^2 \end{bmatrix} \right). \end{aligned}$$

If, on the other hand, the two outputs were generated from the same source (maybe some of the circuitry was shared), then they would undoubtedly be correlated. As we will see shortly, we could model this by letting any of the matrices $\mathbf{A}$, $\mathbf{B}$, or $\mathbf{Q}$ be nondiagonal (see (13.26)). ◇

We now complete our discussion of the vector Gauss-Markov model by deriving its statistical properties. The computations are simple extensions of those for the scalar model. First, we determine an explicit expression for $\mathbf{s}[n]$. From (13.12) we have

$$\begin{aligned} \mathbf{s}[0] &= \mathbf{A}\mathbf{s}[-1] + \mathbf{B}\mathbf{u}[0] \\ \mathbf{s}[1] &= \mathbf{A}\mathbf{s}[0] + \mathbf{B}\mathbf{u}[1] \\ &= \mathbf{A}^2\mathbf{s}[-1] + \mathbf{A}\mathbf{B}\mathbf{u}[0] + \mathbf{B}\mathbf{u}[1] \\ &\text{etc.} \end{aligned}$$

In general, we have

$$\mathbf{s}[n] = \mathbf{A}^{n+1}\mathbf{s}[-1] + \sum_{k=0}^{n} \mathbf{A}^k\mathbf{B}\mathbf{u}[n-k] \tag{13.13}$$

where $\mathbf{A}^0 = \mathbf{I}$. It is seen that $\mathbf{s}[n]$ is a linear function of the initial condition and the driving noise inputs. As a result, $\mathbf{s}[n]$ is a Gaussian random process. It remains only to determine the mean and covariance. From (13.13)

$$\begin{aligned} E(\mathbf{s}[n]) &= \mathbf{A}^{n+1}E(\mathbf{s}[-1]) \\ &= \mathbf{A}^{n+1}\boldsymbol{\mu}_s. \end{aligned} \tag{13.14}$$

The covariance is

$$\begin{aligned}
\mathbf{C}_s[m,n] &= E\left[(\mathbf{s}[m]-E(\mathbf{s}[m]))(\mathbf{s}[n]-E(\mathbf{s}[n]))^T\right] \\
&= E\left[\left(\mathbf{A}^{m+1}(\mathbf{s}[-1]-\boldsymbol{\mu}_s)+\sum_{k=0}^{m}\mathbf{A}^k\mathbf{B}\mathbf{u}[m-k]\right)\right. \\
&\quad \left.\cdot\left(\mathbf{A}^{n+1}(\mathbf{s}[-1]-\boldsymbol{\mu}_s)+\sum_{l=0}^{n}\mathbf{A}^l\mathbf{B}\mathbf{u}[n-l]\right)^T\right] \\
&= \mathbf{A}^{m+1}\mathbf{C}_s\mathbf{A}^{n+1^T}+\sum_{k=0}^{m}\sum_{l=0}^{n}\mathbf{A}^k\mathbf{B}E(\mathbf{u}[m-k]\mathbf{u}^T[n-l])\mathbf{B}^T\mathbf{A}^{l^T}.
\end{aligned}$$

But

$$\begin{aligned}
E(\mathbf{u}[m-k]\mathbf{u}^T[n-l]) &= \mathbf{Q}\delta[l-(n-m+k)] \\
&= \begin{cases} \mathbf{Q} & l=n-m+k \\ \mathbf{0} & \text{otherwise.} \end{cases}
\end{aligned}$$

Thus, for $m \geq n$

$$\mathbf{C}_s[m,n]=\mathbf{A}^{m+1}\mathbf{C}_s\mathbf{A}^{n+1^T}+\sum_{k=m-n}^{m}\mathbf{A}^k\mathbf{B}\mathbf{Q}\mathbf{B}^T\mathbf{A}^{n-m+k^T} \tag{13.15}$$

and for $m < n$

$$\mathbf{C}_s[m,n]=\mathbf{C}_s^T[n,m].$$

The covariance matrix for $\mathbf{s}[n]$ is

$$\begin{aligned}
\mathbf{C}[n] &= \mathbf{C}_s[n,n] \\
&= \mathbf{A}^{n+1}\mathbf{C}_s\mathbf{A}^{n+1^T}+\sum_{k=0}^{n}\mathbf{A}^k\mathbf{B}\mathbf{Q}\mathbf{B}^T\mathbf{A}^{k^T}.
\end{aligned} \tag{13.16}$$

Also, note that the mean and covariance propagation equations can be written as

$$E(\mathbf{s}[n]) = \mathbf{A}E(\mathbf{s}[n-1]) \tag{13.17}$$

$$\mathbf{C}[n] = \mathbf{A}\mathbf{C}[n-1]\mathbf{A}^T+\mathbf{B}\mathbf{Q}\mathbf{B}^T \qquad n \geq 0 \tag{13.18}$$

which follow from (13.12) (see Problem 13.5). As in the scalar case, the covariance matrix decreases due to the $\mathbf{A}\mathbf{C}[n-1]\mathbf{A}^T$ term but increases due to the $\mathbf{B}\mathbf{Q}\mathbf{B}^T$ term. (It can be shown that for a stable process the eigenvalues of $\mathbf{A}$ must be less than 1 in magnitude. See Problem 13.6.) Steady-state properties similar to those of the scalar Gauss-Markov model are evident from (13.14) and (13.16). As $n \to \infty$, the mean will converge to zero or

$$E(\mathbf{s}[n])=\mathbf{A}^{n+1}\boldsymbol{\mu}_s \to \mathbf{0}$$

since the eigenvalues of $\mathbf{A}$ are all less than 1 in magnitude. Also, from (13.16) it can be shown that as $n \to \infty$ (see Problem 13.7)

$$\mathbf{A}^{n+1}\mathbf{C}_s\mathbf{A}^{n+1^T} \to \mathbf{0}$$

so that

$$\mathbf{C}[n] \to \mathbf{C} = \sum_{k=0}^{\infty} \mathbf{A}^k\mathbf{B}\mathbf{Q}\mathbf{B}^T\mathbf{A}^{k^T}. \tag{13.19}$$

It is interesting to note that the steady-state covariance is also the solution of (13.18) if we set $\mathbf{C}[n-1] = \mathbf{C}[n] = \mathbf{C}$ in (13.18). Then, the steady-state covariance satisfies

$$\mathbf{C} = \mathbf{A}\mathbf{C}\mathbf{A}^T + \mathbf{B}\mathbf{Q}\mathbf{B}^T \tag{13.20}$$

and (13.19) is the solution, as can be verified by direct substitution. This is known as the *Lyapunov equation.*

Although in our definition of the Gauss-Markov model we assumed that the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{Q}$ did not depend on n, it is perfectly permissible to do so, and in some cases quite useful. Similar expressions for the mean and covariance can be developed. One major difference though is that the process may not attain a statistical steady-state.

We now summarize the model and its properties.

Theorem 13.1 (Vector Gauss-Markov Model) *The Gauss-Markov model for a $p \times 1$ vector signal $\mathbf{s}[n]$ is*

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] \qquad n \geq 0. \tag{13.21}$$

The $\mathbf{A}$, $\mathbf{B}$ are known matrices having dimensions $p \times p$ and $p \times r$, respectively, and it is assumed that the eigenvalues of $\mathbf{A}$ are less than 1 in magnitude. The driving noise vector $\mathbf{u}[n]$ has dimension $r \times 1$ and is vector WGN or $\mathbf{u}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ with the $\mathbf{u}[n]$'s independent. The initial condition $\mathbf{s}[-1]$ is a $p \times 1$ random vector distributed according to $\mathbf{s}[-1] \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$ and is independent of the $\mathbf{u}[n]$'s. Then, the signal process is Gaussian with mean

$$E(\mathbf{s}[n]) = \mathbf{A}^{n+1}\boldsymbol{\mu}_s \tag{13.22}$$

and covariance for $m \geq n$

$$\begin{aligned} \mathbf{C}_s[m,n] &= E\left[(\mathbf{s}[m] - E(\mathbf{s}[m]))(\mathbf{s}[n] - E(\mathbf{s}[n]))^T\right] \\ &= \mathbf{A}^{m+1}\mathbf{C}_s\mathbf{A}^{n+1^T} + \sum_{k=m-n}^{m} \mathbf{A}^k\mathbf{B}\mathbf{Q}\mathbf{B}^T\mathbf{A}^{n-m+k^T} \end{aligned} \tag{13.23}$$

and for $m < n$

$$\mathbf{C}_s[m,n] = \mathbf{C}_s^T[n,m]$$

and covariance matrix

$$\mathbf{C}[n] = \mathbf{C}_s[n,n] = \mathbf{A}^{n+1}\mathbf{C}_s\mathbf{A}^{n+1^T} + \sum_{k=0}^{n} \mathbf{A}^k\mathbf{B}\mathbf{Q}\mathbf{B}^T\mathbf{A}^{k^T}. \tag{13.24}$$

The mean and covariance propagation equations are

$$E(\mathbf{s}[n]) = \mathbf{A}E(\mathbf{s}[n-1]) \tag{13.25}$$
$$\mathbf{C}[n] = \mathbf{A}\mathbf{C}[n-1]\mathbf{A}^T + \mathbf{B}\mathbf{Q}\mathbf{B}^T. \tag{13.26}$$

13.4 Scalar Kalman Filter

The scalar Gauss-Markov signal model discussed in the previous section had the form

$$s[n] = as[n-1] + u[n] \qquad n \geq 0.$$

We now describe a sequential MMSE estimator which will allow us to estimate $s[n]$ based on the data $\{x[0], x[1], \ldots, x[n]\}$ as n increases. Such an operation is referred to as *filtering*. The approach computes the estimator $\hat{s}[n]$ based on the estimator for the previous time sample $\hat{s}[n-1]$ and so is recursive in nature. This is the so-called *Kalman filter*. As explained in the introduction, the versatility of the Kalman filter accounts for its widespread use. It can be applied to estimation of a scalar Gauss-Markov signal as well as to its vector extension. Furthermore, the data, which previously in all our discussions consisted of a *scalar* sequence such as $\{x[0], x[1], \ldots, x[n]\}$, can be extended to vector observations or $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n]\}$. A common example occurs in array processing in which at each time instant we sample the outputs of a group of sensors. If we have M sensors, then each data sample $\mathbf{x}[n]$ or observation will be a $M \times 1$ vector. Three different levels of generality are now summarized in hierarchical order.

1. scalar state – scalar observation $(s[n-1], x[n])$

2. vector state – scalar observation $(\mathbf{s}[n-1], x[n])$

3. vector state – vector observation $(\mathbf{s}[n-1], \mathbf{x}[n])$.

In this section we discuss the first case, leaving the remaining ones to Section 13.6.

Consider the scalar state equation and the scalar observation equation

$$\begin{aligned} s[n] &= as[n-1] + u[n] \\ x[n] &= s[n] + w[n] \end{aligned} \tag{13.27}$$

where $u[n]$ is zero mean Gaussian noise with independent samples and $E(u^2[n]) = \sigma_u^2$, $w[n]$ is zero mean Gaussian noise with independent samples and $E(w^2[n]) = \sigma_n^2$. We further assume that $s[-1]$, $u[n]$, and $w[n]$ are all independent. Finally, we assume that $s[-1] \sim \mathcal{N}(\mu_s, \sigma_s^2)$. The noise process $w[n]$ differs from WGN only in that its variance is allowed to change with time. To simplify the derivation we will assume that $\mu_s = 0$, so that according to (13.4) $E(s[n]) = 0$ for $n \geq 0$. Later we will account for a nonzero initial signal mean. We wish to estimate $s[n]$ based on the observations $\{x[0], x[1], \ldots, x[n]\}$ or to filter $x[n]$ to produce $\hat{s}[n]$. More generally, the estimator of $s[n]$ based on the observations $\{x[0], x[1], \ldots, x[m]\}$ will be denoted by $\hat{s}[n|m]$. Our criterion of optimality will be the minimum Bayesian MSE or

$$E\left[(s[n] - \hat{s}[n|n])^2\right]$$

where the expectation is with respect to $p(x[0], x[1], \ldots, x[n], s[n])$. But the MMSE estimator is just the mean of the posterior PDF or

$$\hat{s}[n|n] = E(s[n]|x[0], x[1], \ldots, x[n]). \tag{13.28}$$

Using Theorem 10.2 with zero means this becomes

$$\hat{s}[n|n] = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \tag{13.29}$$

since $\theta = s[n]$ and $\mathbf{x} = [x[0]\, x[1] \ldots x[n]]^T$ are jointly Gaussian. Because we are assuming Gaussian statistics for the signal and noise, the MMSE estimator is *linear* and is *identical* in algebraic form to the LMMSE estimator. The algebraic properties allow us to utilize the vector space approach to find the estimator. The implicit linear constraint does not detract from the generality since we already know that the optimal estimator is linear. Furthermore, if the Gaussian assumption is not valid, then the resulting estimator is still valid but can only be said to be the optimal LMMSE estimator. Returning to the sequential computation of (13.29), we note that if $x[n]$ is uncorrelated with $\{x[0], x[1], \ldots, x[n-1]\}$, then from (13.28) and the orthogonality principle we will have (see Example 12.2)

$$\begin{aligned} \hat{s}[n|n] &= E(s[n]|x[0], x[1], \ldots, x[n-1]) + E(s[n]|x[n]) \\ &= \hat{s}[n|n-1] + E(s[n]|x[n]) \end{aligned}$$

which has the desired sequential form. Unfortunately, the $x[n]$'s are correlated due to their dependence on $s[n]$, which is correlated from sample to sample. From our discussions in Chapter 12 of the sequential LMMSE estimator, we can use our vector space interpretation to determine the correction of the old estimator $\hat{s}[n|n-1]$ due to the observation of $x[n]$. Before doing so we will summarize some properties of the MMSE estimator that will be used.

1. The MMSE estimator of θ based on two uncorrelated data vectors, assuming jointly Gaussian statistics, is (see Section 11.4)

$$\begin{aligned} \hat{\theta} &= E(\theta|\mathbf{x}_1, \mathbf{x}_2) \\ &= E(\theta|\mathbf{x}_1) + E(\theta|\mathbf{x}_2) \end{aligned}$$

   if θ is zero mean.

2. The MMSE estimator is additive in that if $\theta = \theta_1 + \theta_2$, then

$$\begin{aligned} \hat{\theta} &= E(\theta|\mathbf{x}) \\ &= E(\theta_1 + \theta_2|\mathbf{x}) \\ &= E(\theta_1|\mathbf{x}) + E(\theta_2|\mathbf{x}). \end{aligned}$$

With these properties we begin the derivation of (13.38)–(13.42). Let $\mathbf{X}[n] = [x[0]\, x[1] \ldots x[n]]^T$ (we now use $\mathbf{X}$ as our notation to avoid confusion with our previous notation

$\mathbf{x}[n]$ which will subsequently be used for vector observations) and $\tilde{x}[n]$ denote the innovation. Recall that the innovation is the part of $x[n]$ that is uncorrelated with the previous samples $\{x[0], x[1], \ldots, x[n-1]\}$ or

$$\tilde{x}[n] = x[n] - \hat{x}[n|n-1]. \tag{13.30}$$

This is because by the orthogonality principle $\hat{x}[n|n-1]$ is the MMSE estimator of $x[n]$ based on the data $\{x[0], x[1], \ldots, x[n-1]\}$, the error or $\tilde{x}[n]$ being orthogonal (uncorrelated) with the data. The data $\mathbf{X}[n-1], \tilde{x}[n]$ are equivalent to the original data set since $x[n]$ may be recovered from

$$\begin{aligned} x[n] &= \tilde{x}[n] + \hat{x}[n|n-1] \\ &= \tilde{x}[n] + \sum_{k=0}^{n-1} a_k x[k] \end{aligned}$$

where the a_k's are the optimal weighting coefficients of the MMSE estimator of $x[n]$ based on $\{x[0], x[1], \ldots, x[n-1]\}$. Now we can rewrite (13.28) as

$$\hat{s}[n|n] = E(s[n]|\mathbf{X}[n-1], \tilde{x}[n])$$

and because $\mathbf{X}[n-1]$ and $\tilde{x}[n]$ are uncorrelated, we have from property 1 that

$$\hat{s}[n|n] = E(s[n]|\mathbf{X}[n-1]) + E(s[n]|\tilde{x}[n]).$$

But $E(s[n]|\mathbf{X}[n-1])$ is the prediction of $s[n]$ based on the previous data, and we denote it by $\hat{s}[n|n-1]$. Explicitly the prediction is, from (13.1) and property 2,

$$\begin{aligned} \hat{s}[n|n-1] &= E(s[n]|\mathbf{X}[n-1]) \\ &= E(as[n-1] + u[n]|\mathbf{X}[n-1]) \\ &= aE(s[n-1]|\mathbf{X}[n-1]) \\ &= a\hat{s}[n-1|n-1] \end{aligned}$$

since $E(u[n]|\mathbf{X}[n-1]) = 0$. This is because

$$E(u[n]|\mathbf{X}[n-1]) = E(u[n]) = 0$$

since $u[n]$ is independent of $\{x[0], x[1], \ldots, x[n-1]\}$. This follows by noting that $u[n]$ is independent of all $w[n]$, and from (13.2) $s[0], s[1], \ldots, s[n-1]$ are linear combinations of the random variables $\{u[0], u[1], \ldots, u[n-1], s[-1]\}$ and these are also independent of $u[n]$. We now have that

$$\hat{s}[n|n] = \hat{s}[n|n-1] + E(s[n]|\tilde{x}[n]) \tag{13.31}$$

where

$$\hat{s}[n|n-1] = a\hat{s}[n-1|n-1].$$

To determine $E(s[n]|\tilde{x}[n])$ we note that it is the MMSE estimator of $s[n]$ based on $\tilde{x}[n]$. As such, it is linear, and because of the zero mean assumption of $s[n]$, it takes the form

$$\begin{aligned} E(s[n]|\tilde{x}[n]) &= K[n]\tilde{x}[n] \\ &= K[n](x[n] - \hat{x}[n|n-1]) \end{aligned}$$

where

$$K[n] = \frac{E(s[n]\tilde{x}[n])}{E(\tilde{x}^2[n])}. \tag{13.32}$$

This follows from the general MMSE estimator for jointly Gaussian θ and x

$$\hat{\theta} = C_{\theta x}C_{xx}^{-1}x = \frac{E(\theta x)}{E(x^2)}x.$$

But $x[n] = s[n] + w[n]$, so that by property 2

$$\begin{aligned} \hat{x}[n|n-1] &= \hat{s}[n|n-1] + \hat{w}[n|n-1] \\ &= \hat{s}[n|n-1] \end{aligned}$$

since $\hat{w}[n|n-1] = 0$ due to $w[n]$ being independent of $\{x[0], x[1], \ldots, x[n-1]\}$. Thus,

$$E(s[n]|\tilde{x}[n]) = K[n](x[n] - \hat{s}[n|n-1])$$

and from (13.31) we now have

$$\hat{s}[n|n] = \hat{s}[n|n-1] + K[n](x[n] - \hat{s}[n|n-1]) \tag{13.33}$$

where

$$\hat{s}[n|n-1] = a\hat{s}[n-1|n-1]. \tag{13.34}$$

It remains only to determine the gain factor $K[n]$. From (13.32) the gain factor is

$$K[n] = \frac{E[s[n](x[n] - \hat{s}[n|n-1])]}{E[(x[n] - \hat{s}[n|n-1])^2]}.$$

To evaluate this we need the results

1.

$$E[s[n](x[n] - \hat{s}[n|n-1])] = E[(s[n] - \hat{s}[n|n-1])(x[n] - \hat{s}[n|n-1])]$$

2.

$$E\left[w[n](s[n] - \hat{s}[n|n-1])\right] = 0.$$

The first result is a consequence of the fact that the innovation

$$\begin{aligned} \tilde{x}[n] &= x[n] - \hat{x}[n|n-1] \\ &= x[n] - \hat{s}[n|n-1] \end{aligned} \tag{13.35}$$

is uncorrelated with the past data and hence with $\hat{s}[n|n-1]$, which is a linear combination of $x[0], x[1], \ldots, x[n-1]$. The second result follows from $s[n]$ and $w[n]$ being uncorrelated and $w[n]$ being uncorrelated with the past data (since $w[n]$ is an uncorrelated process). Using these properties, the gain becomes

$$\begin{aligned} K[n] &= \frac{E[(s[n]-\hat{s}[n|n-1])(x[n]-\hat{s}[n|n-1])]}{E[(s[n]-\hat{s}[n|n-1]+w[n])^2]} \\ &= \frac{E[(s[n]-\hat{s}[n|n-1])^2]}{\sigma_n^2 + E[(s[n]-\hat{s}[n|n-1])^2]}. \end{aligned} \tag{13.36}$$

But the numerator is just the minimum MSE incurred when $s[n]$ is estimated based on the previous data or the minimum one-step prediction error. We will denote this by $M[n|n-1]$, so that

$$K[n] = \frac{M[n|n-1]}{\sigma_n^2 + M[n|n-1]}. \tag{13.37}$$

To evaluate the gain we need an expression for the minimum prediction error. Using (13.34)

$$\begin{aligned} M[n|n-1] &= E[(s[n]-\hat{s}[n|n-1])^2] \\ &= E[(as[n-1]+u[n]-\hat{s}[n|n-1])^2] \\ &= E[(a(s[n-1]-\hat{s}[n-1|n-1])+u[n])^2]. \end{aligned}$$

We note that

$$E[(s[n-1]-\hat{s}[n-1|n-1])u[n]] = 0$$

since $s[n-1]$ depends on $\{u[0], u[1], \ldots, u[n-1], s[-1]\}$, which are independent of $u[n]$, and $\hat{s}[n-1|n-1]$ depends on past data samples or $\{s[0]+w[0], s[1]+w[1], \ldots, s[n-1]+w[n-1]\}$, which also are independent of $u[n]$. Thus,

$$M[n|n-1] = a^2 M[n-1|n-1] + \sigma_u^2.$$

Finally, we require a recursion for $M[n|n]$. Using (13.33), we have

$$\begin{aligned} M[n|n] &= E[(s[n]-\hat{s}[n|n])^2] \\ &= E[(s[n]-\hat{s}[n|n-1]-K[n](x[n]-\hat{s}[n|n-1]))^2] \\ &= E[(s[n]-\hat{s}[n|n-1])^2] - 2K[n]E[(s[n]-\hat{s}[n|n-1])(x[n]-\hat{s}[n|n-1])] \\ &\quad + K^2[n]E[(x[n]-\hat{s}[n|n-1])^2]. \end{aligned}$$

But the first term is $M[n|n-1]$, the second expectation is the numerator of $K[n]$, and the last expectation is the denominator of $K[n]$, as seen from (13.36). Hence, from (13.37)

$$\begin{aligned} M[n|n] &= M[n|n-1] - 2K^2[n](M[n|n-1]+\sigma_n^2) + K[n]M[n|n-1] \\ &= M[n|n-1] - 2K[n]M[n|n-1] + K[n]M[n|n-1] \\ &= (1-K[n])M[n|n-1]. \end{aligned}$$

This completes the derivation of the scalar state–scalar observation Kalman filter. Although tedious, the final equations are actually quite simple and intuitive. We summarize them below. For $n \geq 0$

Prediction:

$$\hat{s}[n|n-1] = a\hat{s}[n-1|n-1]. \tag{13.38}$$

Minimum Prediction MSE:

$$M[n|n-1] = a^2 M[n-1|n-1] + \sigma_u^2. \tag{13.39}$$

Kalman Gain:

$$K[n] = \frac{M[n|n-1]}{\sigma_n^2 + M[n|n-1]}. \tag{13.40}$$

Correction:

$$\hat{s}[n|n] = \hat{s}[n|n-1] + K[n](x[n] - \hat{s}[n|n-1]). \tag{13.41}$$

Minimum MSE:

$$M[n|n] = (1 - K[n])M[n|n-1]. \tag{13.42}$$

Although derived for $\mu_s = 0$ so that $E(s[n]) = 0$, the same equations result for $\mu_s \neq 0$ (see Appendix 13A). Hence, to initialize the equations we use $\hat{s}[-1|-1] = E(s[-1]) = \mu_s$ and $M[-1|-1] = \sigma_s^2$ since this amounts to the estimation of $s[-1]$ without any data. A block diagram of the Kalman filter is given in Figure 13.5. It is interesting to note that the dynamical model for the signal is an integral part of the estimator. Furthermore, we may view the output of the gain block as an estimator $\hat{u}[n]$ of $u[n]$. From (13.38) and (13.41) the signal estimate is

$$\hat{s}[n|n] = a\hat{s}[n-1|n-1] + \hat{u}[n]$$

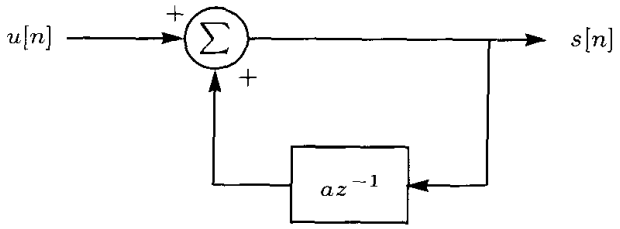
where $\hat{u}[n] = K[n](x[n] - \hat{s}[n|n-1])$. To the extent that this estimate is approximately $u[n]$, we will have $\hat{s}[n|n] \approx s[n]$, as desired. We now consider an example to illustrate the flow of the Kalman filter.

Example 13.2 - Scalar State–Scalar Observation Kalman Filter

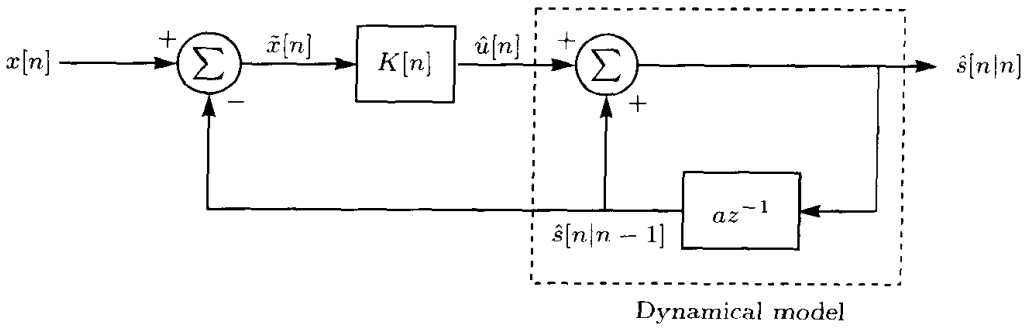
The signal model is a first-order Gauss-Markov process

$$s[n] = \frac{1}{2}s[n-1] + u[n] \qquad n \geq 0$$

where $s[-1] \sim \mathcal{N}(0,1)$ and $\sigma_u^2 = 2$. We assume that the signal is observed in noise $w[n]$, so that the data are $x[n] = s[n] + w[n]$, where $w[n]$ is zero mean Gaussian noise



(a) Dynamical model



(b) Kalman filter

Figure 13.5 Scalar state–scalar observation Kalman filter and relationship to dynamic model

with independent samples, a variance of $\sigma_n^2 = (1/2)^n$, and independent of $s[-1]$ and $u[n]$ for $n \geq 0$. We initialize the filter with

$$\begin{aligned} \hat{s}[-1|-1] &= E(s[-1]) = 0 \\ M[-1|-1] &= E[(s[-1] - \hat{s}[-1|-1])^2] \\ &= E(s^2[-1]) = 1. \end{aligned}$$

According to (13.38) and (13.39), we first predict the $s[0]$ sample to obtain

$$\begin{aligned} \hat{s}[0|-1] &= a\hat{s}[-1|-1] \\ &= \frac{1}{2}(0) = 0 \\ M[0|-1] &= a^2 M[-1|-1] + \sigma_u^2 \\ &= \frac{1}{4}(1) + 2 = \frac{9}{4}. \end{aligned}$$

Next, as $x[0]$ is observed, we correct our predicted estimate by using (13.40) and (13.41)

to yield

$$\begin{aligned} K[0] &= \frac{M[0|-1]}{\sigma_0^2 + M[0|-1]} \\ &= \frac{\frac{9}{4}}{1+\frac{9}{4}} = \frac{9}{13} \end{aligned}$$

$$\begin{aligned} \hat{s}[0|0] &= \hat{s}[0|-1] + K[0](x[0] - \hat{s}[0|-1]) \\ &= 0 + \frac{9}{13}(x[0] - 0) = \frac{9}{13}x[0]. \end{aligned}$$

Next, the minimum MSE is updated using (13.42).

$$\begin{aligned} M[0|0] &= (1 - K[0])M[0|-1] \\ &= \left(1 - \frac{9}{13}\right)\frac{9}{4} = \frac{9}{13}. \end{aligned}$$

For $n = 1$ we obtain the results

$$\begin{aligned} \hat{s}[1|0] &= \frac{9}{26}x[0] \\ M[1|0] &= \frac{113}{52} \\ K[1] &= \frac{113}{129} \\ \hat{s}[1|1] &= \frac{9}{26}x[0] + \frac{113}{129}\left(x[1] - \frac{9}{26}x[0]\right) \\ M[1|1] &= \frac{452}{1677} \end{aligned}$$

which the reader should verify. ◇

We should note that *the same set of equations result if* $E(s[-1]) \neq 0$. In this case the mean of $s[n]$ will be nonzero since $E(s[n]) = a^{n+1}E(s[-1])$ (see (13.3)). This extension is explored in Problem 13.13. Recall that the reason for the zero mean assumption was to allow us to use the orthogonality principle. We now discuss some important properties of the Kalman filter. They are:

1. The Kalman filter extends the sequential MMSE estimator in Chapter 12 to the case where the unknown parameter evolves in time according to the dynamical model. In Chapter 12 we derived the equations for the sequential LMMSE estimator (see (12.47)–(12.49)), which are identical in form to those for the sequential MMSE estimator which assumes Gaussian statistics. In particular, for the scalar

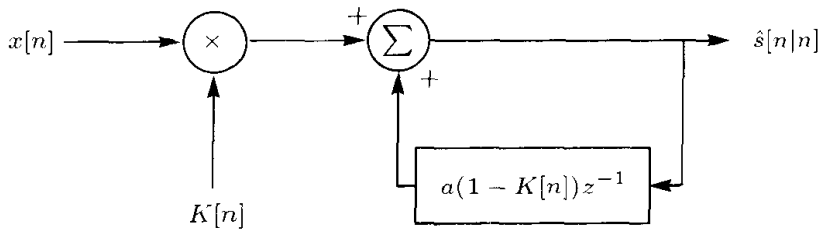


Figure 13.6 Kalman filter as time varying filter

case with $h[n] = 1$ and $\theta[n]$ replaced by $s[n]$, those equations become

$$\hat{s}[n] = \hat{s}[n-1] + K[n](x[n] - \hat{s}[n-1]) \tag{13.43}$$

$$K[n] = \frac{M[n-1]}{\sigma_n^2 + M[n-1]} \tag{13.44}$$

$$M[n] = (1 - K[n])M[n-1]. \tag{13.45}$$

The Kalman filter will reduce to these equations when the parameter to be estimated does not evolve in time. This follows by assuming the driving noise to be zero or $\sigma_u^2 = 0$ and also $a = 1$. Then, the state equation becomes from (13.1) $s[n] = s[n-1]$ or explicitly $s[n] = s[-1] = \theta$. The parameter to be estimated is a constant, which is modeled as the realization of a random variable. Then, the prediction is $\hat{s}[n|n-1] = \hat{s}[n-1|n-1]$ with minimum MSE $M[n|n-1] = M[n-1|n-1]$, so that we can omit the prediction stage of the Kalman filter. This says that the prediction is just the last estimate of $s[n]$. The correction stage reduces to (13.43)–(13.45) since we can let $\hat{s}[n|n] = \hat{s}[n]$, $\hat{s}[n|n-1] = \hat{s}[n-1]$, $M[n|n] = M[n]$, and $M[n|n-1] = M[n-1]$.

2. No matrix inversions are required. This should be compared to the *batch* method of estimating $\theta = s[n]$ as

$$\hat{\theta} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x}$$

where $\mathbf{x} = [x[0]\, x[1] \ldots x[n]]^T$. The use of this formula requires us to invert $\mathbf{C}_{xx}$ for each sample of $s[n]$ to be estimated. And in fact, the dimension of the matrix is $(n+1) \times (n+1)$, becoming larger with n.

3. The Kalman filter is a time varying linear filter. Note from (13.38) and (13.41) that

$$\begin{aligned} \hat{s}[n|n] &= a\hat{s}[n-1|n-1] + K[n](x[n] - a\hat{s}[n-1|n-1]) \\ &= a(1 - K[n])\hat{s}[n-1|n-1] + K[n]x[n]. \end{aligned}$$

This is a first-order recursive filter with time varying coefficients as shown in Figure 13.6.

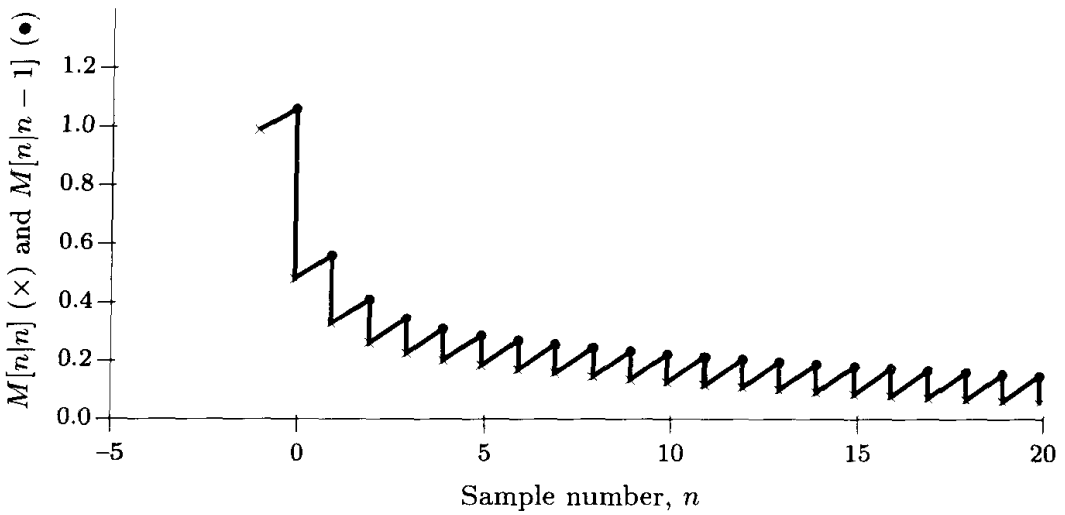


Figure 13.7 Prediction and correction minimum MSE

4. The Kalman filter provides its own performance measure. From (13.42) the minimum Bayesian MSE is computed as an integral part of the estimator. Also, the error measure may be computed off-line, i.e., before any data are collected. This is because $M[n|n]$ depends only on (13.39) and (13.40), which are independent of the data. We will see later that for the extended Kalman filter the minimum MSE sequence must be computed on-line (see Section 13.7).

5. The prediction stage increases the error, while the correction stage decreases it. There is an interesting interplay between the minimum prediction MSE and the minimum MSE. If as $n \to \infty$ a steady-state condition is achieved, then $M[n|n]$ and $M[n|n-1]$ each become constant with $M[n|n-1] > M[n-1|n-1]$ (see Problem 13.14). Hence, the error will increase after the prediction stage. When the new data sample is obtained, we correct the estimate, which decreases the MSE according to (13.42) (since $K[n] < 1$). An example of this is shown in Figure 13.7, in which $a = 0.99$, $\sigma_u^2 = 0.1$, $\sigma_n^2 = 0.9^{n+1}$, and $M[-1|-1] = 1$. We will say more about this in the next section when we discuss the relationship of the Kalman filter to the Wiener filter.

6. Prediction is an integral part of the Kalman filter. It is seen from (13.38) that to determine the best filtered estimate of $s[n]$ we employ predictions. We can find the best one-step prediction of $s[n]$ based on $\{x[0], x[1], \ldots, x[n-1]\}$ from (13.38). If we desire the best two-step prediction, we can obtain it easily by noting that this is the best estimate of $s[n+1]$ based on $\{x[0], x[1], \ldots, x[n-1]\}$. To find this we let $\sigma_n^2 \to \infty$, implying that $x[n]$ is so noisy that the Kalman filter will not use it. Then, $\hat{s}[n+1|n-1]$ is just the optimal two-step prediction. To evaluate this we have from (13.38)

$$\hat{s}[n+1|n] = a\hat{s}[n|n]$$

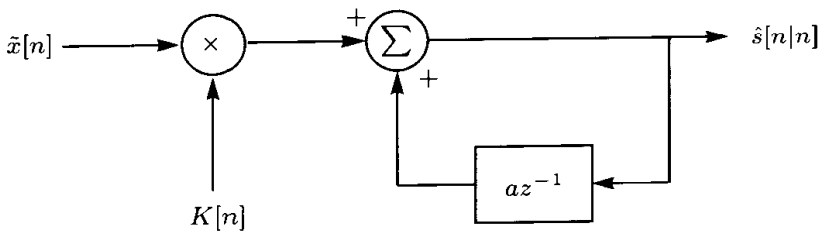


Figure 13.8 Innovation-driven Kalman filter

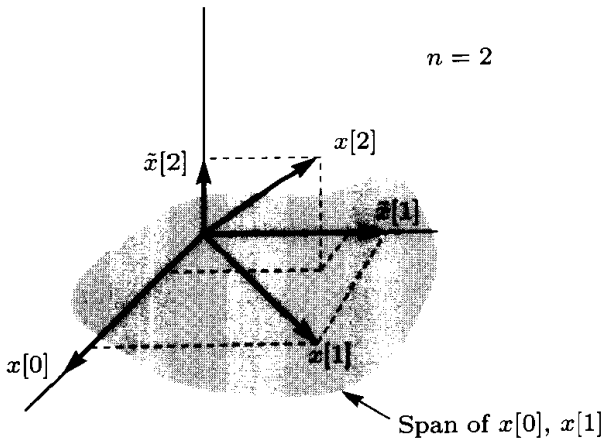


Figure 13.9 Orthogonality (uncorrelated property) of innovation sequence $\tilde{x}(n)$

and since $K[n] \to 0$, we have from (13.41) that $\hat{s}[n|n] = \hat{s}[n|n-1]$, where $\hat{s}[n|n-1] = a\hat{s}[n-1|n-1]$. Thus, the optimal two-step prediction is

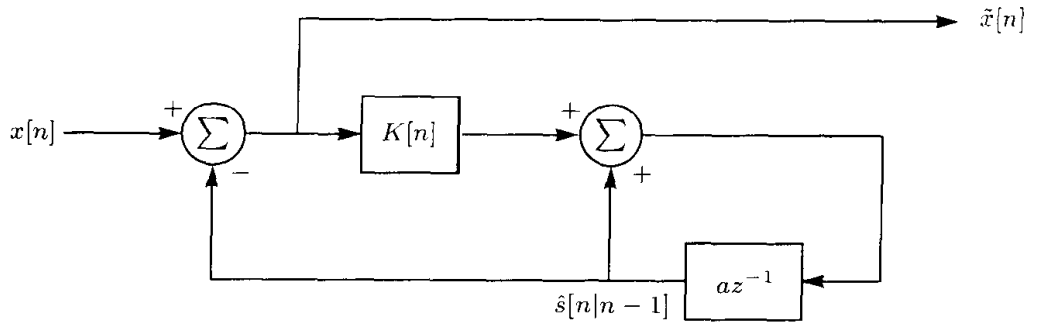
$$\begin{aligned} \hat{s}[n+1|n-1] &= \hat{s}[n+1|n] \\ &= a\hat{s}[n|n] \\ &= a\hat{s}[n|n-1] \\ &= a^2\hat{s}[n-1|n-1]. \end{aligned}$$

This can be generalized to the l-step predictor, as shown in Problem 13.15.

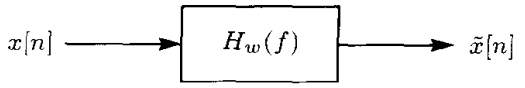
7. The Kalman filter is driven by the uncorrelated innovation sequence and in steady-state can also be viewed as a whitening filter. Note from (13.38) and (13.41) that

$$\hat{s}[n|n] = a\hat{s}[n-1|n-1] + K[n](x[n] - \hat{s}[n|n-1])$$

so that the input to the Kalman filter is the innovation sequence $\tilde{x}[n] = x[n] - \hat{s}[n|n-1]$ (see (13.35)) as shown in Figure 13.8. We know from our discussion of the vector space approach that $\tilde{x}[n]$ is uncorrelated with $\{x[0], x[1], \ldots, x[n-1]\}$, which translates into a sequence of uncorrelated random variables as shown in Figure 13.9. Alternatively, if we view $\tilde{x}[n]$ as the Kalman filter *output* and if the filter attains steady-state, then it becomes a linear time invariant whitening filter as shown in Figure 13.10. This is discussed in detail in the next section.



(a) Kalman filter



(b) System model

Figure 13.10 Whitening filter interpretation of Kalman filter

8. The Kalman filter is optimal in that it minimizes the Bayesian MSE for each estimator $\hat{s}[n]$. If the Gaussian assumption is not valid, then it is still the optimal *linear* MMSE estimator as described in Chapter 12.

All these properties of the Kalman filter carry over to the vector state case, except for property 2, if the observations are also vectors.

13.5 Kalman Versus Wiener Filters

The causal infinite length Wiener filter described in Chapter 12 produced an estimate of $s[n]$ as the output of a linear time invariant filter or

$$\hat{s}[n] = \sum_{k=0}^{\infty} h[k]x[n-k].$$

The estimator of $s[n]$ is based on the present data sample and the infinite past. To determine the filter impulse response $h[k]$ analytically we needed to assume that the signal $s[n]$ and noise $w[n]$ were WSS processes, so that the Wiener-Hopf equations could be solved (see Problem 12.16). In the Kalman filter formulation the signal and noise need not be WSS. The variance of $w[n]$ may change with n, and furthermore, $s[n]$ will only be WSS as $n \to \infty$. Additionally, the Kalman filter produces estimates based on only the data samples from 0 to n, not the infinite past as assumed by the infinite length Wiener filter. The two filters will, however, be the same as $n \to \infty$ if $\sigma_n^2 = \sigma^2$. This is because $s[n]$ will approach statistical steady-state as shown in Section 13.3, i.e., it will become an AR(1) process, and the estimator will then be based on the present

and the infinite past. Since the Kalman filter will approach a linear time invariant filter, we can let $K[n] \rightarrow K[\infty]$, $M[n|n] \rightarrow M[\infty]$, and $M[n|n-1] \rightarrow M_p[\infty]$, where $M_p[\infty]$ is the steady-state one-step prediction error. To find the steady-state Kalman filter we need to first find $M[\infty]$. From (13.39), (13.40), and (13.42) we have

$$\begin{aligned} M[\infty] &= \left(1 - \frac{M_p[\infty]}{\sigma^2 + M_p[\infty]}\right) M_p[\infty] \\ &= \frac{\sigma^2 M_p[\infty]}{M_p[\infty] + \sigma^2} \\ &= \frac{\sigma^2(a^2 M[\infty] + \sigma_u^2)}{a^2 M[\infty] + \sigma_u^2 + \sigma^2} \end{aligned} \tag{13.46}$$

which must be solved for $M[\infty]$. The resulting equation is termed the steady-state *Ricatti equation* and is seen to be quadratic. Once $M[\infty]$ has been found, $M_p[\infty]$ can be determined and finally $K[\infty]$. Then, the steady-state Kalman filter takes the form of the first-order recursive filter shown in Figure 13.6 with $K[n]$ replaced by the constant $K[\infty]$. Note that a simple way of solving the Ricatti equation numerically is to run the Kalman filter until it converges. This will produce the desired time invariant filter. The steady-state filter will have the form

$$\begin{aligned} \hat{s}[n|n] &= a\hat{s}[n-1|n-1] + K[\infty](x[n] - a\hat{s}[n-1|n-1]) \\ &= a(1 - K[\infty])\hat{s}[n-1|n-1] + K[\infty]x[n] \end{aligned}$$

so that its steady-state transfer function is

$$\mathcal{H}_\infty(z) = \frac{K[\infty]}{1 - a(1 - K[\infty])z^{-1}}. \tag{13.47}$$

As an example, if $a = 0.9$, $\sigma_u^2 = 1$, $\sigma^2 = 1$, we will find from (13.46) that $M[\infty] = 0.5974$ (the other solution is negative). Hence, from (13.39) $M_p[\infty] = 1.4839$, and from (13.40) $K[\infty] = 0.5974$. The steady-state frequency response is

$$\begin{aligned} H_\infty(f) &= \mathcal{H}_\infty(\exp(j2\pi f)) \\ &= \frac{0.5974}{1 - 0.3623\exp(-j2\pi f)} \end{aligned}$$

whose magnitude is shown in Figure 13.11 as a solid line versus the PSD of the steady-state signal

$$\begin{aligned} P_{ss}(f) &= \frac{\sigma_u^2}{|1 - a\exp(-j2\pi f)|^2} \\ &= \frac{1}{|1 - 0.9\exp(-j2\pi f)|^2} \end{aligned}$$

shown as a dashed line. The same results would be obtained if the Wiener-Hopf equation had been solved for the causal infinite length Wiener filter. Hence, the steady-state

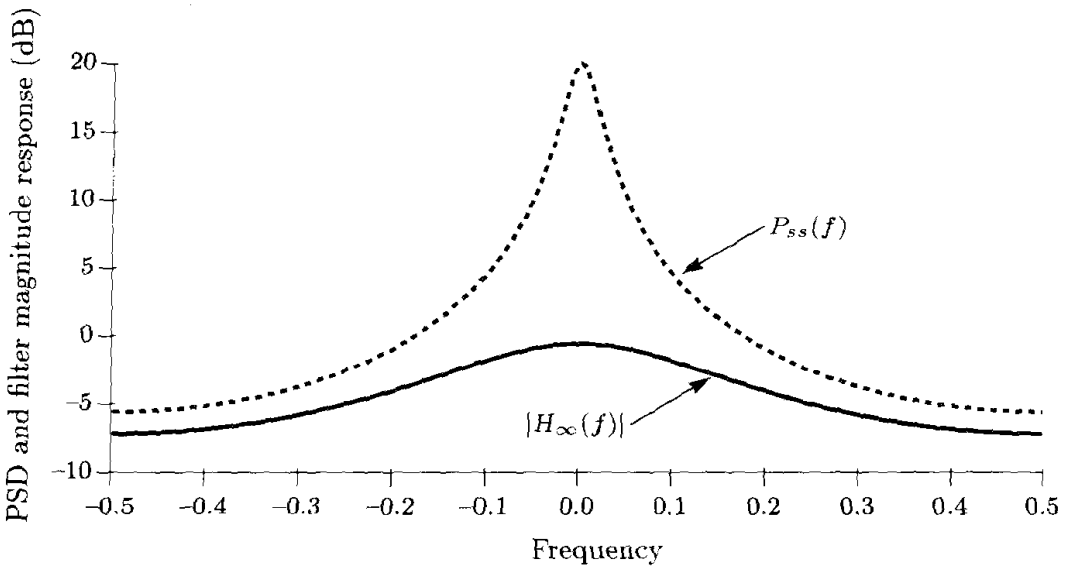


Figure 13.11 Signal PSD and steady-state Kalman filter magnitude response

Kalman filter is equivalent to the causal infinite length Wiener filter if the signal and the noise become WSS as $n \to \infty$.

Finally, the whitening filter property discussed in property 7 in the previous section may be verified for this example. The innovation is

$$\begin{aligned} \tilde{x}[n] &= x[n] - \hat{s}[n|n-1] \\ &= x[n] - a\hat{s}[n-1|n-1]. \end{aligned} \tag{13.48}$$

But in steady-state $\hat{s}[n|n]$ is the output of the filter with system function $\mathcal{H}_\infty(z)$ driven by $x[n]$. Thus, the system function relating the input $x[n]$ to the output $\tilde{x}[n]$ is, from (13.48) and (13.47),

$$\begin{aligned} \mathcal{H}_w(z) &= 1 - az^{-1}\mathcal{H}_\infty(z) \\ &= 1 - \frac{az^{-1}K[\infty]}{1 - a(1 - K[\infty])z^{-1}} \\ &= \frac{1 - az^{-1}}{1 - a(1 - K[\infty])z^{-1}}. \end{aligned}$$

For this example we have the whitening filter frequency response

$$H_w(f) = \frac{1 - 0.9\exp(-j2\pi f)}{1 - 0.3623\exp(-j2\pi f)}$$

whose magnitude is plotted in Figure 13.12 as a solid line along with the PSD of $x[n]$,

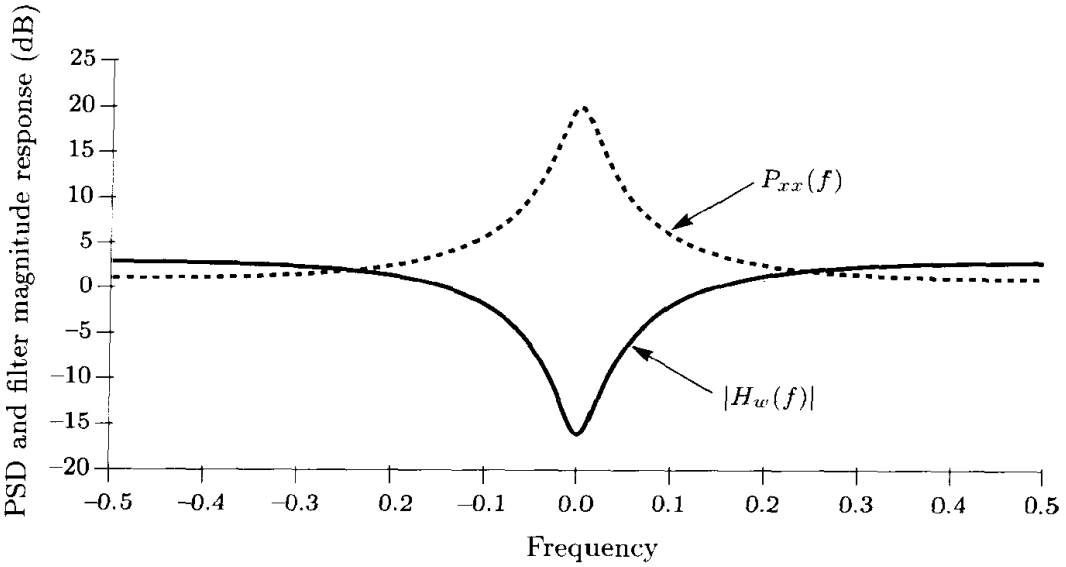


Figure 13.12 Whitening filter property of steady-state Kalman filter

shown as a dashed line. Note that the PSD of $x[n]$ is

$$\begin{aligned} P_{xx}(f) &= P_{ss}(f) + \sigma^2 \\ &= \frac{\sigma_u^2}{|1 - a\exp(-j2\pi f)|^2} + \sigma^2 \\ &= \frac{\sigma_u^2 + \sigma^2|1 - a\exp(-j2\pi f)|^2}{|1 - a\exp(-j2\pi f)|^2} \end{aligned}$$

which for this example is

$$P_{xx}(f) = \frac{1 + |1 - 0.9\exp(-j2\pi f)|^2}{|1 - 0.9\exp(-j2\pi f)|^2}.$$

The PSD at the output of the whitening filter is therefore

$$|H_w(f)|^2 P_{xx}(f) = \frac{1 + |1 - 0.9\exp(-j2\pi f)|^2}{|1 - 0.3623\exp(-j2\pi f)|^2}$$

which can be verified to be the constant PSD $P_{\tilde{x}\tilde{x}}(f) = 2.48$. In general, we have

$$P_{\tilde{x}\tilde{x}}(f) = \frac{\sigma_{\tilde{x}}^2}{|H_w(f)|^2}$$

where $\sigma_{\tilde{x}}^2$ is the variance of the innovation. Thus, as shown in Figure 13.13, the output PSD of a filter with frequency response $H_w(f)$ is a flat PSD with height $\sigma_{\tilde{x}}^2$.

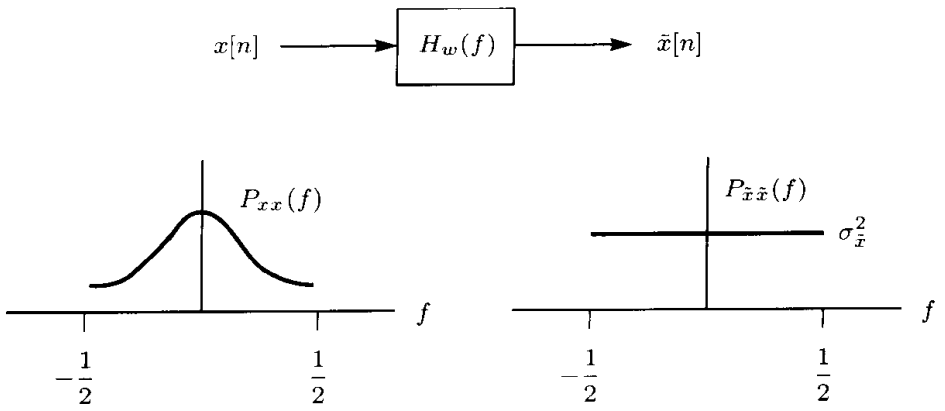


Figure 13.13 Input and output PSDs of steady-state Kalman whitening filter

13.6 Vector Kalman Filter

The scalar state–scalar observation Kalman filter is easily generalized. The two generalizations are to replace $s[n]$ by $\mathbf{s}[n]$, where $\mathbf{s}[n]$ obeys the Gauss-Markov model described in Theorem 13.1, and to replace the scalar observation $x[n]$ by the vector observation $\mathbf{x}[n]$. The first generalization will produce the vector state–scalar observation Kalman filter, while the second leads to the most general form, the vector state–vector observation Kalman filter. In either case the state model is, from Theorem 13.1,

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] \qquad n \geq 0$$

where $\mathbf{A}, \mathbf{B}$ are known $p \times p$ and $p \times r$ matrices, $\mathbf{u}[n]$ is vector WGN with $\mathbf{u}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$, $\mathbf{s}[-1] \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$, and $\mathbf{s}[-1]$ is independent of the $\mathbf{u}[n]$'s. The vector state–scalar observation Kalman filter assumes that the observations follow the Bayesian linear model (see Section 10.6) with the added assumption that the noise covariance matrix is *diagonal*. Thus, for the nth data sample we have

$$x[n] = \mathbf{h}^T[n]\mathbf{s}[n] + w[n] \tag{13.49}$$

where $\mathbf{h}[n]$ is a known $p \times 1$ vector and $w[n]$ is zero mean Gaussian noise with uncorrelated samples, with variance σ_n^2, and also independent of $\mathbf{s}[-1]$ and $\mathbf{u}[n]$. The data model of (13.49) is called the *observation or measurement equation*. An example is given in Section 13.8, where we wish to track the coefficients of a random time varying FIR filter. For that example, the state is comprised of the impulse response values. The Kalman filter for this setup is derived in exactly the same manner as for the scalar state case. The derivation is included in Appendix 13A. We now summarize the results. The reader should note the similarity to (13.38)–(13.42).

Prediction:

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{A}\hat{\mathbf{s}}[n-1|n-1]. \tag{13.50}$$

Minimum Prediction MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n-1] = \mathbf{A}\mathbf{M}[n-1|n-1]\mathbf{A}^T + \mathbf{B}\mathbf{Q}\mathbf{B}^T. \tag{13.51}$$

Kalman Gain Vector ($p \times 1$):

$$\mathbf{K}[n] = \frac{\mathbf{M}[n|n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{M}[n|n-1]\mathbf{h}[n]}. \tag{13.52}$$

Correction:

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](x[n] - \mathbf{h}^T[n]\hat{\mathbf{s}}[n|n-1]). \tag{13.53}$$

Minimum MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n])\mathbf{M}[n|n-1] \tag{13.54}$$

where the mean square error matrices are defined as

$$\mathbf{M}[n|n] = E\left[(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n])(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n])^T\right] \tag{13.55}$$

$$\mathbf{M}[n|n-1] = E\left[(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1])(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1])^T\right]. \tag{13.56}$$

To initialize the equations we use $\hat{\mathbf{s}}[-1|-1] = E(\mathbf{s}[-1]) = \boldsymbol{\mu}_s$ and $\mathbf{M}[-1|-1] = \mathbf{C}_s$. The order of the recursion is identical to the scalar state–scalar observation case. Also, as before, no matrix inversions are required, but this is no longer true when we consider the vector observation case. If $\mathbf{A} = \mathbf{I}$ and $\mathbf{B} = \mathbf{0}$, then the equations are identical to the sequential LMMSE estimator (see (12.47)–(12.49)). This is because the signal models are identical, the signal being constant in time. An application example is given in Section 13.8.

Finally, we generalize the Kalman filter to the case of *vector observations*, which is quite common in practice. An example is in array processing, where at each time instant we sample the output of an array of M sensors. Then, the observations are $\mathbf{x}[n] = [x_1[n]\, x_2[n] \ldots x_M[n]]^T$, where $x_i[n]$ is the output of the ith sensor at time n. The observations are modeled using the Bayesian linear model

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where *each* vector observation has this form. Indexing the quantities by n and replacing $\boldsymbol{\theta}$ by $\mathbf{s}$, we have as our observation model

$$\mathbf{x}[n] = \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] \tag{13.57}$$

where $\mathbf{H}[n]$ is a known $M \times p$ matrix, $\mathbf{x}[n]$ is an $M \times 1$ observation vector, and $\mathbf{w}[n]$ is a $M \times 1$ *observation* noise sequence. The $\mathbf{w}[n]$'s are independent of each other and of $\mathbf{u}[n]$ and $\mathbf{s}[-1]$, and $\mathbf{w}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C}[n])$. Except for the dependence of the covariance matrix on n, $\mathbf{w}[n]$ can be thought of as vector WGN. For the array processing problem $\mathbf{s}[n]$ represents a vector of p transmitted signals, which are modeled as random,

and $\mathbf{H}[n]$ models the linear transformation due to the medium. The medium may be modeled as time varying since $\mathbf{H}[n]$ depends on n. Hence, $\mathbf{H}[n]\mathbf{s}[n]$ is the signal output at the M sensors. Also, the sensor outputs are corrupted by noise $\mathbf{w}[n]$. The statistical assumptions on $\mathbf{w}[n]$ indicate the noise is correlated from sensor to sensor at the same time instant with covariance $\mathbf{C}[n]$, and this correlation varies with time. From time instant to time instant, however, the noise samples are independent since $E(\mathbf{w}[i]\mathbf{w}^T[j]) = \mathbf{0}$ for $i \neq j$. With this data model we can derive the vector state–vector observation Kalman filter, the most general estimator. Because of the large number of assumptions required, we summarize the results as a theorem.

Theorem 13.2 (Vector Kalman Filter) *The $p \times 1$ signal vector $\mathbf{s}[n]$ evolves in time according to the Gauss-Markov model*

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] \qquad n \geq 0$$

where $\mathbf{A}, \mathbf{B}$ are known matrices of dimension $p \times p$ and $p \times r$, respectively. The driving noise vector $\mathbf{u}[n]$ has the PDF $\mathbf{u}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ and is independent from sample to sample, so that $E(\mathbf{u}[m]\mathbf{u}^T[n]) = \mathbf{0}$ for $m \neq n$ ($\mathbf{u}[n]$ is vector WGN). The initial state vector $\mathbf{s}[-1]$ has the PDF $\mathbf{s}[-1] \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$ and is independent of $\mathbf{u}[n]$.

The $M \times 1$ observation vectors $\mathbf{x}[n]$ are modeled by the Bayesian linear model

$$\mathbf{x}[n] = \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] \qquad n \geq 0$$

where $\mathbf{H}[n]$ is a known $M \times p$ observation matrix (which may be time varying) and $\mathbf{w}[n]$ is an $M \times 1$ observation noise vector with PDF $\mathbf{w}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C}[n])$ and is independent from sample to sample, so that $E(\mathbf{w}[m]\mathbf{w}^T[n]) = \mathbf{0}$ for $m \neq n$. (If $\mathbf{C}[n]$ did not depend on n, then $\mathbf{w}[n]$ would be vector WGN.)

The MMSE estimator of $\mathbf{s}[n]$ based on $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n]\}$ or

$$\hat{\mathbf{s}}[n|n] = E(\mathbf{s}[n]|\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n])$$

can be computed sequentially in time using the following recursion:

Prediction:

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{A}\hat{\mathbf{s}}[n-1|n-1]. \tag{13.58}$$

Minimum Prediction MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n-1] = \mathbf{A}\mathbf{M}[n-1|n-1]\mathbf{A}^T + \mathbf{B}\mathbf{Q}\mathbf{B}^T. \tag{13.59}$$

Kalman Gain Matrix ($p \times M$):

$$\mathbf{K}[n] = \mathbf{M}[n|n-1]\mathbf{H}^T[n]\left(\mathbf{C}[n] + \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n]\right)^{-1}. \tag{13.60}$$

Correction:

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]). \tag{13.61}$$

Minimum MSE Matrix *($p \times p$):*

$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1]. \tag{13.62}$$

The recursion is initialized by $\hat{\mathbf{s}}[-1|-1] = \boldsymbol{\mu}_s$, *and* $\mathbf{M}[-1|-1] = \mathbf{C}_s$.

All the comments of the scalar state–scalar observation Kalman filter apply here as well, with the exception of the need for matrix inversions. We now require the inversion of an $M \times M$ matrix to find the Kalman gain. If the dimension of the state vector p is less than the dimension of the observation vector M, a more efficient implementation of the Kalman filter can be obtained. Referred to as the *information form*, this alternative Kalman filter is described in [Anderson and Moore 1979].

Before concluding the discussion of linear Kalman filters, it is worthwhile to note that the Kalman filter summarized in the previous theorem is still not the most general one. It is, however, adequate for many practical problems. Extensions can be made by letting the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{Q}$ be time varying. Fortuitously, the equations that result are identical to those of the previous theorem when we replace $\mathbf{A}$ by $\mathbf{A}[n]$, $\mathbf{B}$ by $\mathbf{B}[n]$, and $\mathbf{Q}$ by $\mathbf{Q}[n]$. Also, it is possible to extend the results to colored observation noise and to signal models with deterministic inputs (in addition to the driving noise). Finally, smoothing equations have also been derived based on the Kalman philosophy. These extensions are described in [Anderson and Moore 1979, Gelb 1974, Mendel 1987].

13.7 Extended Kalman Filter

In practice we are often faced with a state equation and/or an observation equation which is nonlinear. The previous approaches then are no longer valid. A simple example that will be explored in some detail in Section 13.8 is vehicle tracking. For this problem the observations or measurements are range estimates $\hat{R}[n]$ and bearing estimates $\hat{\beta}[n]$. If the vehicle state is the position (r_x, r_y) in Cartesian coordinates (it is assumed to travel in the x-y plane), then the noiseless measurements are related to the unknown parameters by

$$\begin{aligned} R[n] &= \sqrt{r_x^2[n] + r_y^2[n]} \\ \beta[n] &= \arctan \frac{r_y[n]}{r_x[n]}. \end{aligned}$$

Due to measurement errors, however, we obtain the estimates $\hat{R}[n]$ and $\hat{\beta}[n]$, which are assumed to be the true range and bearing plus measurement noise. Hence, we have for our measurements

$$\begin{aligned} \hat{R}[n] &= R[n] + w_R[n] \\ \hat{\beta}[n] &= \beta[n] + w_\beta[n] \end{aligned} \tag{13.63}$$

or

$$\begin{aligned}\hat{R}[n] &= \sqrt{r_x^2[n] + r_y^2[n]} + w_R[n] \\ \hat{\beta}[n] &= \arctan \frac{r_y[n]}{r_x[n]} + w_\beta[n].\end{aligned}$$

Clearly, we *cannot* express these in the linear model form or as

$$\mathbf{x}[n] = \mathbf{H}[n]\boldsymbol{\theta}[n] + \mathbf{w}[n]$$

where $\boldsymbol{\theta}[n] = [r_x[n]\, r_y[n]]^T$. The observation equation is *nonlinear*.

An example of a nonlinear state equation occurs if we assume that the vehicle is traveling in a given direction at a known fixed speed and we choose polar coordinates, range and bearing, to describe the state. (Note that this choice would render the measurement equation linear as described by (13.63)). Then, ignoring the driving noise, the state equation becomes

$$\begin{aligned}r_x[n] &= v_x n\Delta + r_x[0] \\ r_y[n] &= v_y n\Delta + r_y[0]\end{aligned} \tag{13.64}$$

where (v_x, v_y) is the known velocity, Δ is the time interval between samples, and $(r_x[0], r_y[0])$ is the initial position. This can be expressed alternatively as

$$\begin{aligned}r_x[n] &= r_x[n-1] + v_x\Delta \\ r_y[n] &= r_y[n-1] + v_y\Delta\end{aligned}$$

so that the range becomes

$$\begin{aligned}R[n] &= \sqrt{r_x^2[n-1] + r_y^2[n-1] + 2v_x\Delta r_x[n-1] + 2v_y\Delta r_y[n-1] + (v_x^2 + v_y^2)\Delta^2} \\ &= \sqrt{R^2[n-1] + 2R[n-1]\Delta(v_x\cos\beta[n-1] + v_y\sin\beta[n-1]) + (v_x^2 + v_y^2)\Delta^2}.\end{aligned}$$

This is clearly very nonlinear in range and bearing. In general, we may be faced with sequential state estimation where the state and/or observation equations are nonlinear. Then, instead of our linear Kalman filter models

$$\begin{aligned}\mathbf{s}[n] &= \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] \\ \mathbf{x}[n] &= \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n]\end{aligned}$$

we would have

$$\mathbf{s}[n] = \mathbf{a}(\mathbf{s}[n-1]) + \mathbf{B}\mathbf{u}[n] \tag{13.65}$$

$$\mathbf{x}[n] = \mathbf{h}(\mathbf{s}[n]) + \mathbf{w}[n] \tag{13.66}$$

where $\mathbf{a}$ is a p-dimensional function and $\mathbf{h}$ is an M-dimensional function. The dimensions of the remaining matrices and vectors are the same as before. Now $\mathbf{a}(\mathbf{s}[n-1])$

represents the true physical model for the evolution of the state, while $\mathbf{u}[n]$ accounts for the modeling errors, unforeseen inputs, etc. Likewise, $\mathbf{h}(\mathbf{s}[n])$ represents the transformation from the state variables to the ideal observations (without noise). For this case the MMSE estimator is intractable. The only hope is an approximate solution based on linearizing $\mathbf{a}$ and $\mathbf{h}$, much the same as was done for nonlinear LS, where the data were nonlinearly related to the unknown parameters. The result of this linearization and the subsequent application of the *linear* Kalman filter of (13.58)–(13.62) results in the *extended Kalman filter*. It has *no optimality properties*, and its performance will depend on the accuracy of the linearization. Being a *dynamic* linearization there is no way to determine its performance beforehand.

Proceeding with the derivation, we linearize $\mathbf{a}(\mathbf{s}[n-1])$ about the estimate of $\mathbf{s}[n-1]$ or about $\hat{\mathbf{s}}[n-1|n-1]$. Likewise, we linearize $\mathbf{h}(\mathbf{s}[n])$ about the estimate of $\mathbf{s}[n]$ based on the previous data or $\hat{\mathbf{s}}[n|n-1]$ since from (13.61) we will need the linearized observation equation to determine $\hat{\mathbf{s}}[n|n]$. Hence, a first-order Taylor expansion yields

$$\begin{aligned}
\mathbf{a}(\mathbf{s}[n-1]) &\approx \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) \\
&\quad + \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} (\mathbf{s}[n-1] - \hat{\mathbf{s}}[n-1|n-1]) \\
\mathbf{h}(\mathbf{s}[n]) &\approx \mathbf{h}(\hat{\mathbf{s}}[n|n-1]) + \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]} (\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1]).
\end{aligned}$$

We let the Jacobians be denoted by

$$\begin{aligned}
\mathbf{A}[n-1] &= \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} \\
\mathbf{H}[n] &= \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]}
\end{aligned}$$

so that the linearized state and observation equations become from (13.65) and (13.66)

$$\begin{aligned}
\mathbf{s}[n] &= \mathbf{A}[n-1]\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] + (\mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) - \mathbf{A}[n-1]\hat{\mathbf{s}}[n-1|n-1]) \\
\mathbf{x}[n] &= \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] + (\mathbf{h}(\hat{\mathbf{s}}[n|n-1]) - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]).
\end{aligned}$$

The equations differ from our standard ones in that $\mathbf{A}$ is now time varying and both equations have *known* terms added to them. It is shown in Appendix 13B that the linear Kalman filter for this model, which is the *extended* Kalman filter, is

Prediction:

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]). \tag{13.67}$$

Minimum Prediction MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n-1] = \mathbf{A}[n-1]\mathbf{M}[n-1|n-1]\mathbf{A}^T[n-1] + \mathbf{B}\mathbf{Q}\mathbf{B}^T. \tag{13.68}$$

Kalman Gain Matrix ($p \times M$):

$$\mathbf{K}[n] = \mathbf{M}[n|n-1]\mathbf{H}^T[n]\left(\mathbf{C}[n] + \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n]\right)^{-1}. \tag{13.69}$$

Correction:

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{\mathbf{s}}[n|n-1])). \tag{13.70}$$

Minimum MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1] \tag{13.71}$$

where

$$\begin{aligned} \mathbf{A}[n-1] &= \left.\frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]}\right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} \\ \mathbf{H}[n] &= \left.\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]}\right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]} \end{aligned}$$

Note that in contrast to the linear Kalman filter the gain and MSE matrices must be computed on-line, as they depend upon the state estimates via $\mathbf{A}[n-1]$ and $\mathbf{H}[n]$. Also, the use of the term MSE matrix is itself a misnomer since the MMSE estimator has not been implemented but only an approximation to it. In the next section we will apply the extended Kalman filter to vehicle tracking.

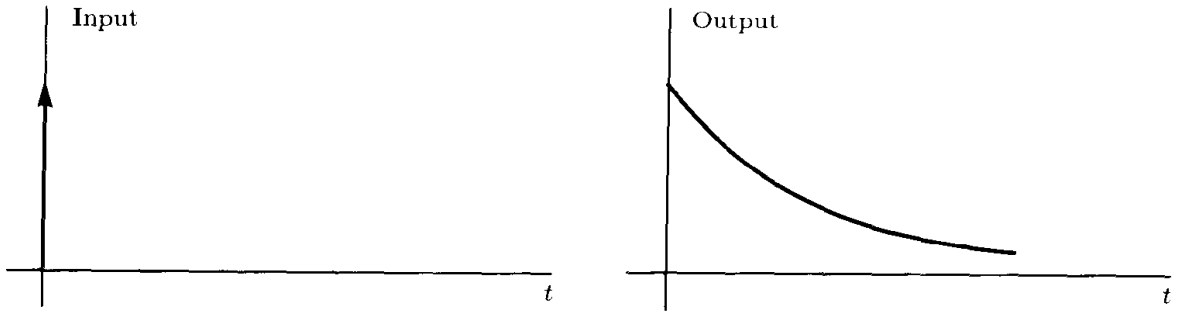
13.8 Signal Processing Examples

We now examine some common signal processing applications of the linear and extended Kalman filters.

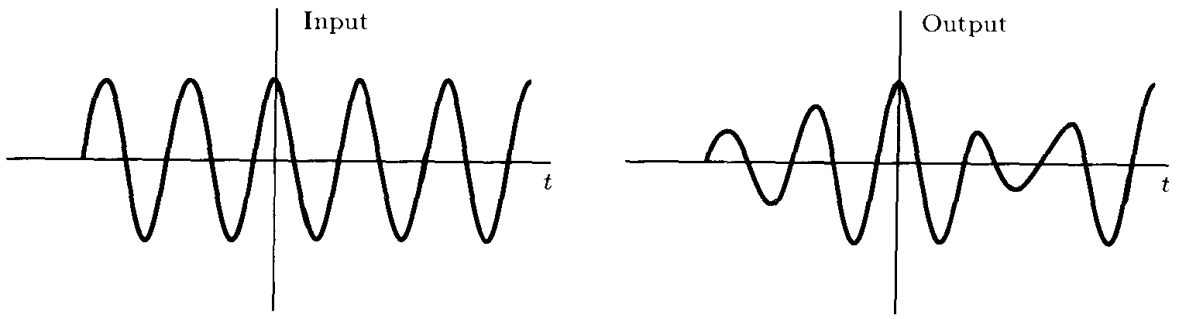
Example 13.3 - Time Varying Channel Estimation

Many transmission channels can be characterized as being linear but *not* time invariant. These are referred to by various names such as *fading dispersive* channels or *fading multipath* channels. They arise in communication problems in which the troposphere is used as a medium or in sonar in which the ocean is used [Kennedy 1969]. In either case, the medium acts as a linear filter, causing an impulse at the input to appear as a continuous waveform at the output (the dispersive or multipath nature), as shown in Figure 13.14a. This effect is the result of a continuum of propagation paths, i.e., multipath, each of which delays and attenuates the input signal. Additionally, however, a sinusoid at the input will appear as a narrowband signal at the output or one whose amplitude is modulated (the fading nature), as shown in Figure 13.14b. This effect is due to the changing character of the medium, for example, the movement of the scatterers. A little thought will convince the reader that the channel is acting as a linear time varying filter. If we sample the output of the channel, then it can be shown that a good model is the low-pass tapped delay line model as shown in Figure 13.15 [Van Trees 1971]. The input-output description of this system is

$$y[n] = \sum_{k=0}^{p-1} h_n[k]v[n-k]. \tag{13.72}$$



(a) Multipath channel



(b) Fading channel

Figure 13.14 Input-output waveforms for fading and multipath channels

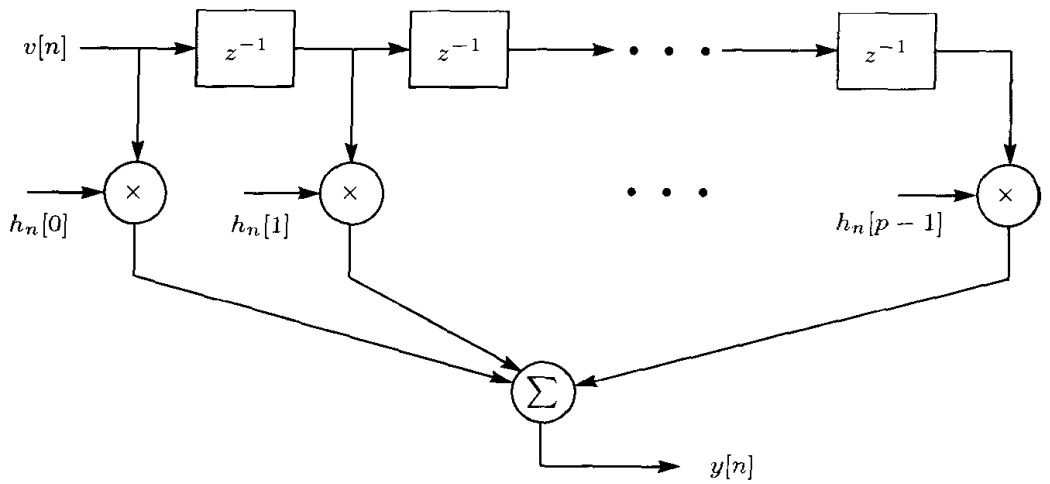


Figure 13.15 Tapped delay line channel model

This is really nothing more than an FIR filter with time-varying coefficients. To design effective communication or sonar systems it is necessary to have knowledge of these coefficients. Hence, the problem becomes one of estimating $h_n[k]$ based on the noise corrupted output of the channel

$$x[n] = \sum_{k=0}^{p-1} h_n[k]v[n-k] + w[n] \tag{13.73}$$

where $w[n]$ is observation noise. A similar problem was addressed in Example 4.3, except that there we assumed the filter coefficients to be time invariant. Consequently, the linear model could be applied to estimate the deterministic parameters. It is not possible to extend that approach to our current problem since there are too many parameters to estimate. To see why this is so we let $p = 2$ and assume that $v[n] = 0$ for $n < 0$. The observations are, from (13.73),

$$\begin{aligned} x[0] &= h_0[0]v[0] + h_0[1]v[-1] + w[0] = h_0[0]v[0] + w[0] \\ x[1] &= h_1[0]v[1] + h_1[1]v[0] + w[1] \\ x[2] &= h_2[0]v[2] + h_2[1]v[1] + w[2] \\ &\text{etc.} \end{aligned}$$

It is seen that for $n \geq 1$ we have two new parameters for each new data sample. Even without corrupting noise we cannot determine the tapped delay line weights. A way out of this problem is to realize that the weights will not change rapidly from sample to sample, as for example, in a slow-fading channel. For example, in Figure 13.14b this would correspond to an amplitude modulation (caused by the time variation of the weights) which is slow. Statistically, we may interpret the slow variation as a high degree of correlation between samples of the same tap weight. This observation naturally leads us to model the tap weights as random variables whose time variation is described by a Gauss-Markov model. The use of such a signal model allows us to fix the correlation between the successive values of a given tap weight in time. Hence, we suppose that the state vector is

$$\mathbf{h}[n] = \mathbf{A}\mathbf{h}[n-1] + \mathbf{u}[n]$$

where $\mathbf{h}[n] = [h_n[0]\, h_n[1] \ldots h_n[p-1]]^T$, $\mathbf{A}$ is a known $p \times p$ matrix, and $\mathbf{u}[n]$ is vector WGN with covariance matrix $\mathbf{Q}$. (The reader should note that $\mathbf{h}[n]$ no longer refers to the observation vector as in (13.49) but is now the signal $\mathbf{s}[n]$.) A standard assumption that is made to simplify the modeling is that of *uncorrelated scattering* [Van Trees 1971]. It assumes that the tap weights are uncorrelated with each other and hence independent due to the jointly Gaussian assumption. As a result, we can let $\mathbf{A}$, $\mathbf{Q}$, and $\mathbf{C}_h$, the covariance matrix of $\mathbf{h}[-1]$, be diagonal matrices. The vector Gauss-Markov model then becomes p independent scalar models. The measurement model is, from (13.73),

$$x[n] = \underbrace{\begin{bmatrix} v[n] & v[n-1] & \ldots & v[n-p+1] \end{bmatrix}}_{\mathbf{v}^T[n]} \mathbf{h}[n] + w[n]$$

where $w[n]$ is assumed to be WGN with variance σ^2 and the $v[n]$ sequence is assumed known (since we provide the input to the channel). We can now form the MMSE estimator for the tapped delay line weights recursively in time using the Kalman filter equations for a vector state and scalar observations. With obvious changes in notation we have from (13.50)–(13.54)

$$\begin{aligned}
\hat{\mathbf{h}}[n|n-1] &= \mathbf{A}\hat{\mathbf{h}}[n-1|n-1] \\
\mathbf{M}[n|n-1] &= \mathbf{A}\mathbf{M}[n-1|n-1]\mathbf{A}^T + \mathbf{Q} \\
\mathbf{K}[n] &= \frac{\mathbf{M}[n|n-1]\mathbf{v}[n]}{\sigma^2 + \mathbf{v}^T[n]\mathbf{M}[n|n-1]\mathbf{v}[n]} \\
\hat{\mathbf{h}}[n|n] &= \hat{\mathbf{h}}[n|n-1] + \mathbf{K}[n](x[n] - \mathbf{v}^T[n]\hat{\mathbf{h}}[n|n-1]) \\
\mathbf{M}[n|n] &= (\mathbf{I} - \mathbf{K}[n]\mathbf{v}^T[n])\mathbf{M}[n|n-1]
\end{aligned}$$

and is initialized by $\hat{\mathbf{h}}[-1|-1] = \boldsymbol{\mu}_h$, $\mathbf{M}[-1|-1] = \mathbf{C}_h$. As an example, we now implement the Kalman filter estimator for a tapped delay line having $p = 2$ weights. We assume a state model with

$$\begin{aligned}
\mathbf{A} &= \begin{bmatrix} 0.99 & 0 \\ 0 & 0.999 \end{bmatrix} \\
\mathbf{Q} &= \begin{bmatrix} 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}.
\end{aligned}$$

A particular realization is shown in Figure 13.16, in which $h_n[0]$ is decaying to zero while $h_n[1]$ is fairly constant. This is because the mean of the weights will be zero in steady-state (see (13.4)). Due to the smaller value of $[\mathbf{A}]_{11}$, $h_n[0]$ will decay more rapidly. Also, note that the eigenvalues of $\mathbf{A}$ are just the diagonal elements and they are less than 1 in magnitude. For this tap weight realization and the input shown in Figure 13.17a, the output is shown in Figure 13.17b as determined from (13.72). When observation noise is added with $\sigma^2 = 0.1$, we have the channel output shown in Figure 13.17c. We next apply the Kalman filter with $\hat{\mathbf{h}}[-1|-1] = \mathbf{0}$ and $\mathbf{M}[-1|-1] = 100\mathbf{I}$, which were chosen to reflect little knowledge about the initial state. In the theoretical development of the Kalman filter the initial state estimate is given by the mean of $\mathbf{s}[-1]$. In practice this is seldom known, so that we usually just choose an arbitrary initial state estimate with a large initial MSE matrix to avoid "biasing" the Kalman filter towards that assumed state. The estimated tap weights are shown in Figure 13.18. After an initial transient the Kalman filter "locks on" to the true weights and tracks them closely. The Kalman filter gains are shown in Figure 13.19. They appear to attain a periodic steady-state, although this behavior is different than the usual steady-state discussed previously since $\mathbf{v}[n]$ varies with time and so true steady-state is never attained. Also, at times the gain is zero, as for example in $[\mathbf{K}]_1 = K_1[n]$ for $0 \leq n \leq 4$. This is because at these times $\mathbf{v}[n]$ is zero due to the zero input and thus the observations contain only noise. The Kalman filter ignores these data samples by forcing the gain to be zero. Finally, the minimum MSEs are shown in Figure 13.20 and are seen to decrease monotonically, although this generally will not be the case for a Kalman filter. ◇

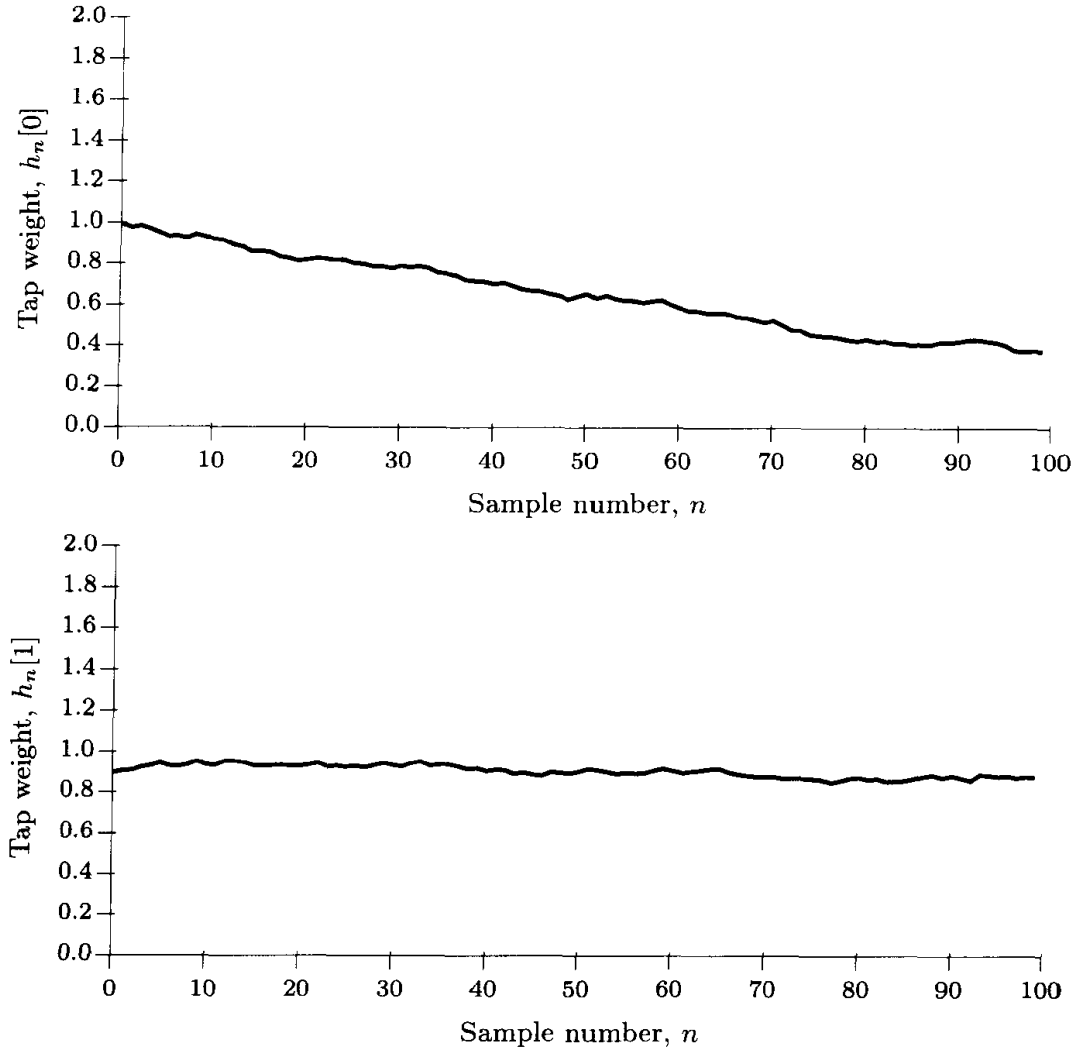


Figure 13.16 Realization of TDL coefficients

Example 13.4 - Vehicle Tracking

In this example we use an extended Kalman filter to track the position and velocity of a vehicle moving in a nominal given direction and at a nominal speed. The measurements are noisy versions of the range and bearing. Such a track is shown in Figure 13.21. In arriving at a model for the dynamics of the vehicle we assume a constant velocity, perturbed only by wind gusts, slight speed corrections, etc., as might occur in an aircraft. We model these perturbations as noise inputs, so that the velocity components in the x and y directions at time n are

$$\begin{aligned} v_x[n] &= v_x[n-1] + u_x[n] \\ v_y[n] &= v_y[n-1] + u_y[n]. \end{aligned} \tag{13.74}$$

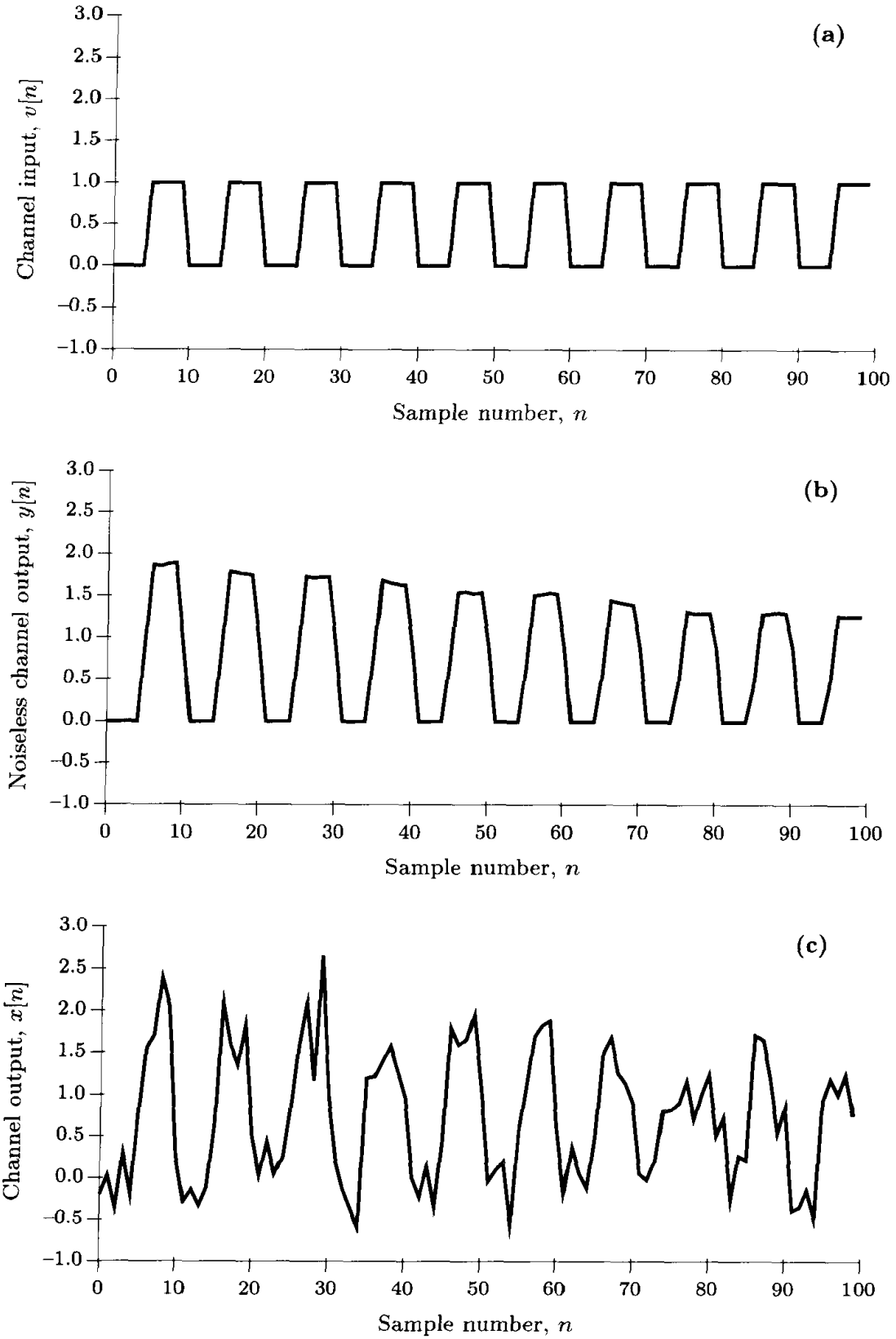


Figure 13.17 Input-output waveforms of channel

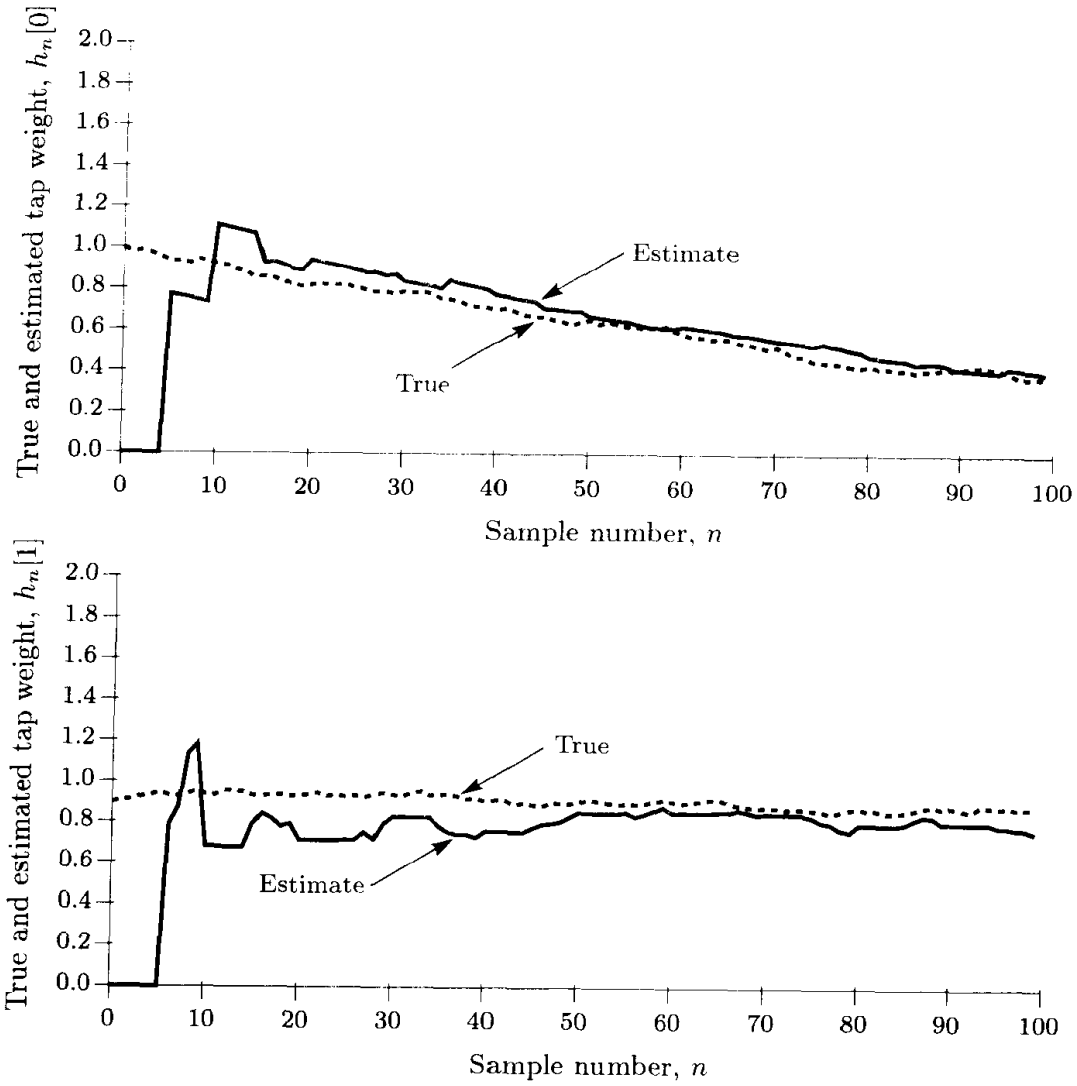


Figure 13.18 Kalman filter estimate

Without the noise perturbations $u_x[n], u_y[n]$ the velocities would be constant, and hence the vehicle would be modeled as traveling in a straight line as indicated by the dashed line in Figure 13.21. From the equations of motion the position at time n is

$$\begin{aligned} r_x[n] &= r_x[n-1] + v_x[n-1]\Delta \\ r_y[n] &= r_y[n-1] + v_y[n-1]\Delta \end{aligned} \tag{13.75}$$

where Δ is the time interval between samples. In this discretized model of the equations of motion the vehicle is modeled as moving at the velocity of the previous time instant and then changing abruptly at the next time instant, an approximation to the true continuous behavior. Now, we choose the signal vector as consisting of the position and

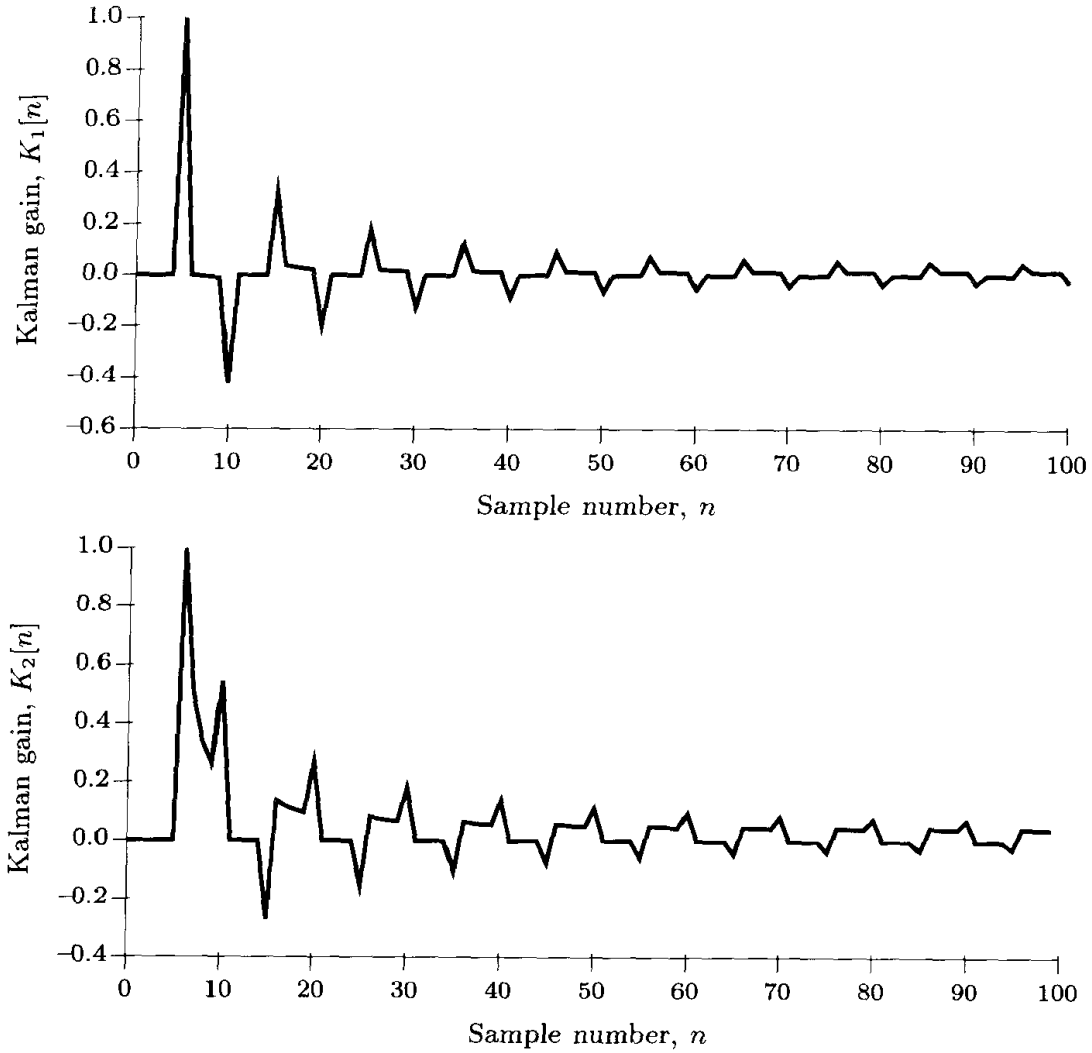


Figure 13.19 Kalman filter gains

velocity components or

$$\mathbf{s}[n] = \begin{bmatrix} r_x[n] \\ r_y[n] \\ v_x[n] \\ v_y[n] \end{bmatrix}$$

and from (13.74) and (13.75) it is seen to satisfy

$$\underbrace{\begin{bmatrix} r_x[n] \\ r_y[n] \\ v_x[n] \\ v_y[n] \end{bmatrix}}_{\mathbf{s}[n]} = \underbrace{\begin{bmatrix} 1 & 0 & \Delta & 0 \\ 0 & 1 & 0 & \Delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} r_x[n-1] \\ r_y[n-1] \\ v_x[n-1] \\ v_y[n-1] \end{bmatrix}}_{\mathbf{s}[n-1]} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ u_x[n] \\ u_y[n] \end{bmatrix}}_{\mathbf{u}[n]}. \tag{13.76}$$

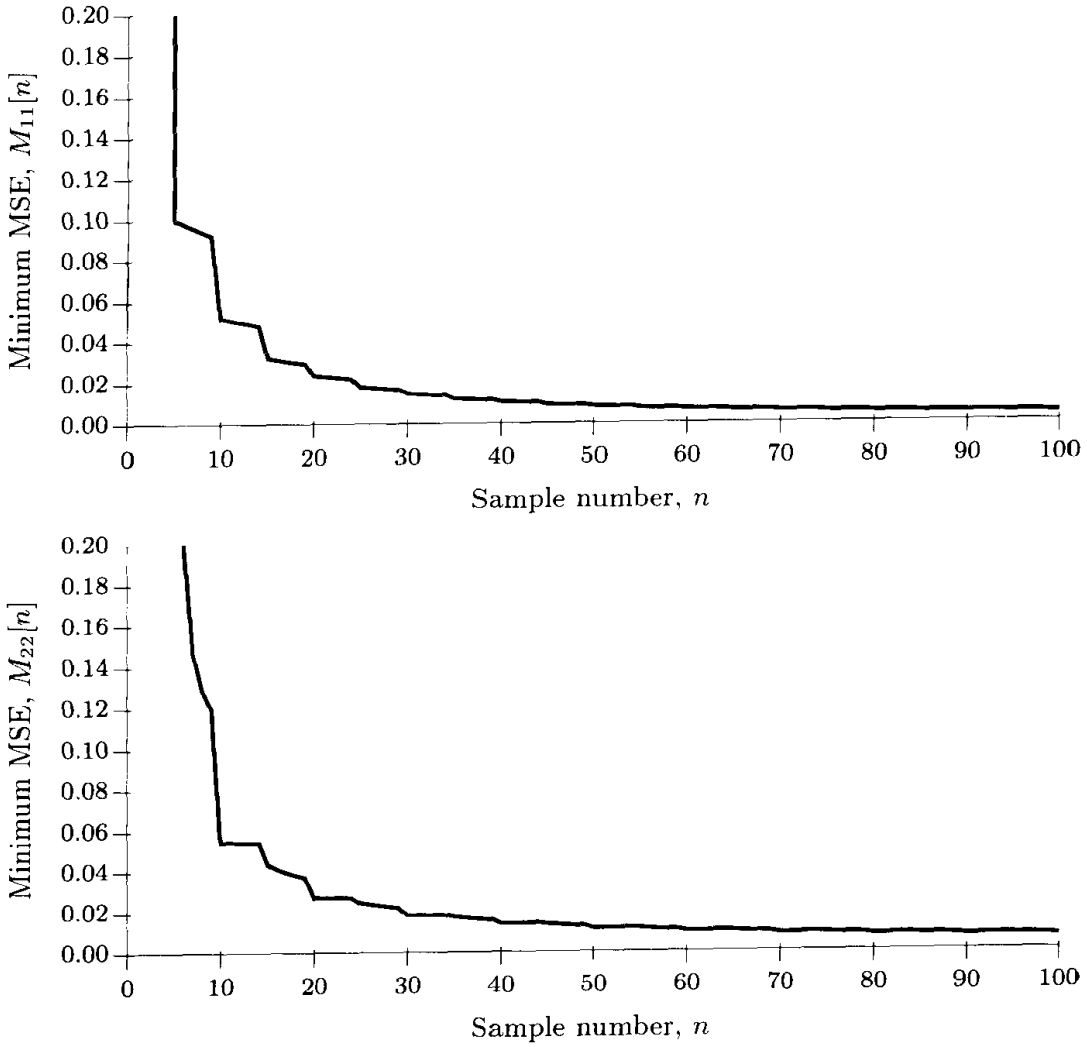


Figure 13.20 Kalman filter minimum MSE

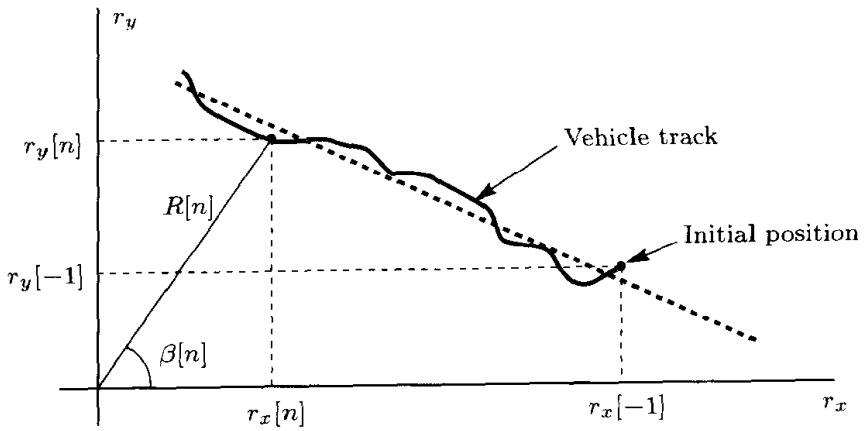


Figure 13.21 Typical track of vehicle moving in given direction at constant speed

The measurements are noisy observations of the range and bearing

$$\begin{aligned} R[n] &= \sqrt{r_x^2[n] + r_y^2[n]} \\ \beta[n] &= \arctan \frac{r_y[n]}{r_x[n]} \end{aligned}$$

or

$$\begin{aligned} \hat{R}[n] &= R[n] + w_R[n] \\ \hat{\beta}[n] &= \beta[n] + w_\beta[n]. \end{aligned} \tag{13.77}$$

In general terms the observation equation of (13.77) is

$$\mathbf{x}[n] = \mathbf{h}(\mathbf{s}[n]) + \mathbf{w}[n]$$

where $\mathbf{h}$ is the function

$$\mathbf{h}(\mathbf{s}[n]) = \begin{bmatrix} \sqrt{r_x^2[n] + r_y^2[n]} \\ \arctan \dfrac{r_y[n]}{r_x[n]} \end{bmatrix}.$$

Unfortunately, the measurement vector is nonlinear in the signal parameters. To estimate the signal vector we will need to apply an extended Kalman filter (see (13.67)–(13.71)). Since the state equation of (13.76) is linear, we need only determine

$$\mathbf{H}[n] = \left. \frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]} \right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]}$$

because $\mathbf{A}[n]$ is just $\mathbf{A}$ as given in (13.76). Differentiating the observation equation, we have the Jacobian

$$\frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]} = \begin{bmatrix} \dfrac{r_x[n]}{R[n]} & \dfrac{r_y[n]}{R[n]} & 0 & 0 \\ \dfrac{-r_y[n]}{R^2[n]} & \dfrac{r_x[n]}{R^2[n]} & 0 & 0 \end{bmatrix}.$$

Finally, we need to specify the covariances of the driving noise and observation noise. If we assume that the wind gusts, speed corrections, etc., are just as likely to occur in any direction and with the same magnitude, then it seems reasonable to assign the same variances to $u_x[n]$ and $u_y[n]$ and to assume that they are independent. Call the common variance σ_u^2. Then, we have

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_u^2 & 0 \\ 0 & 0 & 0 & \sigma_u^2 \end{bmatrix}.$$

The exact value to use for σ_u^2 should depend on the possible change in the velocity component from sample to sample since $u_x[n] = v_x[n] - v_x[n-1]$. This is just the

acceleration times Δ and should be derivable from the physics of the vehicle. In specifying the variances of the measurement noise we note that the measurement error can be thought of as the estimation error of $\hat{R}[n]$ and $\hat{\beta}[n]$ as seen from (13.77). We usually assume the estimation errors $w_R[n], w_\beta[n]$ to be zero mean. Then, the variance of $w_R[n]$, for example, is $E(w_R^2[n]) = E[(\hat{R}[n] - R[n])^2]$. This variance is sometimes derivable but in most instances is not. One possibility is to assume that $E(w_R^2[n])$ does not depend on the PDF of $R[n]$, so that $E(w_R^2[n]) = E[(\hat{R}[n] - R[n])^2 | R[n]]$. Equivalently, we could regard $R[n]$ as a *deterministic* parameter so that the variance of $w_R[n]$ is just the classical estimator variance. As such, if $\hat{R}[n]$ were the MLE, then assuming long data records and/or high SNRs, we could assume that the variance attains the CRLB. Using this approach, we could then make use of the CRLB for range and bearing such as was derived in Examples 3.13 and 3.15 to set the variances. For simplicity we usually assume the estimation errors to be independent and the variances to be time invariant (although this is not always valid). Hence, we have

$$\mathbf{C}[n] = \mathbf{C} = \begin{bmatrix} \sigma_R^2 & 0 \\ 0 & \sigma_\beta^2 \end{bmatrix}.$$

In summary, the extended Kalman filter equations for this problem are, from (13.67)–(13.71),

$$\begin{aligned}
\hat{\mathbf{s}}[n|n-1] &= \mathbf{A}\hat{\mathbf{s}}[n-1|n-1] \\
\mathbf{M}[n|n-1] &= \mathbf{A}\mathbf{M}[n-1|n-1]\mathbf{A}^T + \mathbf{Q} \\
\mathbf{K}[n] &= \mathbf{M}[n|n-1]\mathbf{H}^T[n]\left(\mathbf{C} + \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n]\right)^{-1} \\
\hat{\mathbf{s}}[n|n] &= \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{\mathbf{s}}[n|n-1])) \\
\mathbf{M}[n|n] &= (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1]
\end{aligned}$$

where

$$\begin{aligned}
\mathbf{A} &= \begin{bmatrix} 1 & 0 & \Delta & 0 \\ 0 & 1 & 0 & \Delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
\mathbf{Q} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_u^2 & 0 \\ 0 & 0 & 0 & \sigma_u^2 \end{bmatrix} \\
\mathbf{x}[n] &= \begin{bmatrix} \hat{R}[n] \\ \hat{\beta}[n] \end{bmatrix} \\
\mathbf{h}(\mathbf{s}[n]) &= \begin{bmatrix} \sqrt{r_x^2[n] + r_y^2[n]} \\ \arctan \dfrac{r_y[n]}{r_x[n]} \end{bmatrix}
\end{aligned}$$

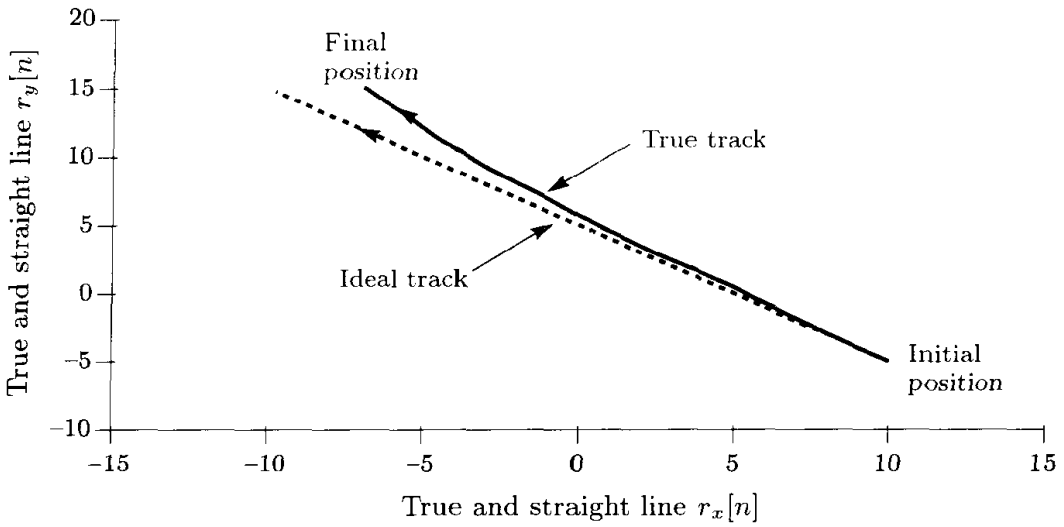


Figure 13.22 Realization of vehicle track

$$\mathbf{H}[n] = \left[\begin{array}{cccc} \dfrac{r_x[n]}{\sqrt{r_x^2[n]+r_y^2[n]}} & \dfrac{r_y[n]}{\sqrt{r_x^2[n]+r_y^2[n]}} & 0 & 0 \\ \dfrac{-r_y[n]}{r_x^2[n]+r_y^2[n]} & \dfrac{r_x[n]}{r_x^2[n]+r_y^2[n]} & 0 & 0 \end{array} \right] \Bigg|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]}$$

$$\mathbf{C} = \left[\begin{array}{cc} \sigma_R^2 & 0 \\ 0 & \sigma_\beta^2 \end{array} \right]$$

and the initial conditions are $\hat{\mathbf{s}}[-1|-1] = \boldsymbol{\mu}_s$, $\mathbf{M}[-1|-1] = \mathbf{C}_s$. As an example, consider the ideal straight line trajectory shown in Figure 13.22 as a dashed line. The coordinates are given by

$$\begin{aligned} r_x[n] &= 10 - 0.2n \\ r_y[n] &= -5 + 0.2n \end{aligned}$$

for $n = 0, 1, \ldots, 100$, where we have assumed $\Delta = 1$ for convenience. From (13.75) this trajectory assumes $v_x = -0.2, v_y = 0.2$. To accommodate a more realistic vehicle track we introduce driving or plant noise, so that the vehicle state is described by (13.76) with a driving noise variance of $\sigma_u^2 = 0.0001$. With an initial state of

$$\mathbf{s}[-1] = \left[\begin{array}{c} 10 \\ -5 \\ -0.2 \\ 0.2 \end{array} \right], \tag{13.78}$$

which is identical to that of the initial state of the straight line trajectory, a realization of the vehicle position $[r_x[n]\, r_y[n]]^T$ is shown in Figure 13.22 as the solid curve. The state equation of (13.76) has been used to generate the realization. Note that as time

increases, the true trajectory gradually deviates from the straight line. It can be shown that the variances of $v_x[n]$ and $v_y[n]$ will eventually increase to infinity (see Problem 13.22), causing $r_x[n]$ and $r_y[n]$ to quickly become unbounded. Thus, this modeling is valid for only a portion of the trajectory. The true range and bearing are shown in Figure 13.23. We assume the measurement noise variances to be $\sigma_R^2 = 0.1$ and $\sigma_\beta^2 = 0.01$, where β is measured in radians. In Figure 13.24 we compare the true trajectory with the noise corrupted one as obtained from

$$\begin{aligned} \hat{r}_x[n] &= \hat{R}[n]\cos\hat{\beta}[n] \\ \hat{r}_y[n] &= \hat{R}[n]\sin\hat{\beta}[n]. \end{aligned}$$

To employ an extended Kalman filter we must specify an initial state estimate. In practice, it is unlikely that we will have knowledge of the position and speed. Thus, for the sake of illustration we choose an initial state that is quite far from the true one or

$$\hat{\mathbf{s}}[-1|-1] = \begin{bmatrix} 5 \\ 5 \\ 0 \\ 0 \end{bmatrix}$$

and so as not to "bias" the extended Kalman filter we assume a large initial MSE or $\mathbf{M}[-1|-1] = 100\mathbf{I}$. The results of an extended Kalman filter are shown in Figure 13.25 as the solid curve. Initially, because of the poor state estimate, the error is large. This is also reflected in the MSE curves in Figure 13.26 (actually these are only estimates based on our linearization). However, after about 20 samples the extended Kalman filter attains the track. Also, it is interesting to note that the minimum MSE does not monotonically decrease as it did in the previous example. On the contrary, it increases for part of the time. This is explained by contrasting the Kalman filter with the sequential LMMSE estimator in Chapter 12. For the latter we estimated the same parameter as we received more and more data. Consequently, the minimum MSE decreased or at worst remained constant. Here, however, each time we receive a new data sample we are estimating a new parameter. The increased uncertainty of the new parameter due to the influence of the driving noise input may be large enough to offset the knowledge gained by observing a new data sample, causing the minimum MSE to increase (see Problem 13.23). As a final remark, in this simulation the extended Kalman filter appeared to be quite tolerant of linearization errors due to a poor initial state estimate. In general, however, we cannot expect to be so fortunate. ◇

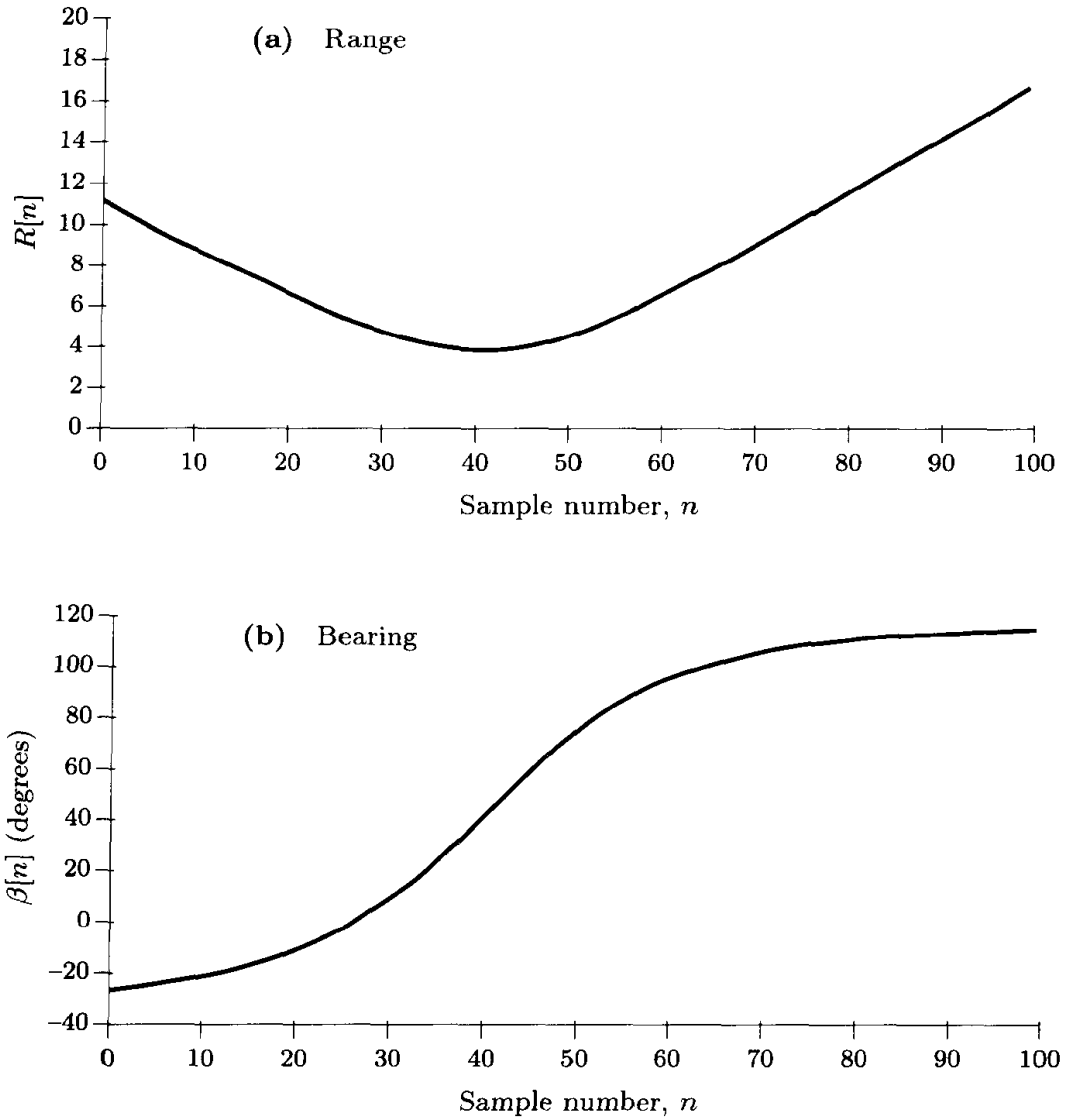


Figure 13.23 Range and bearing of true vehicle track

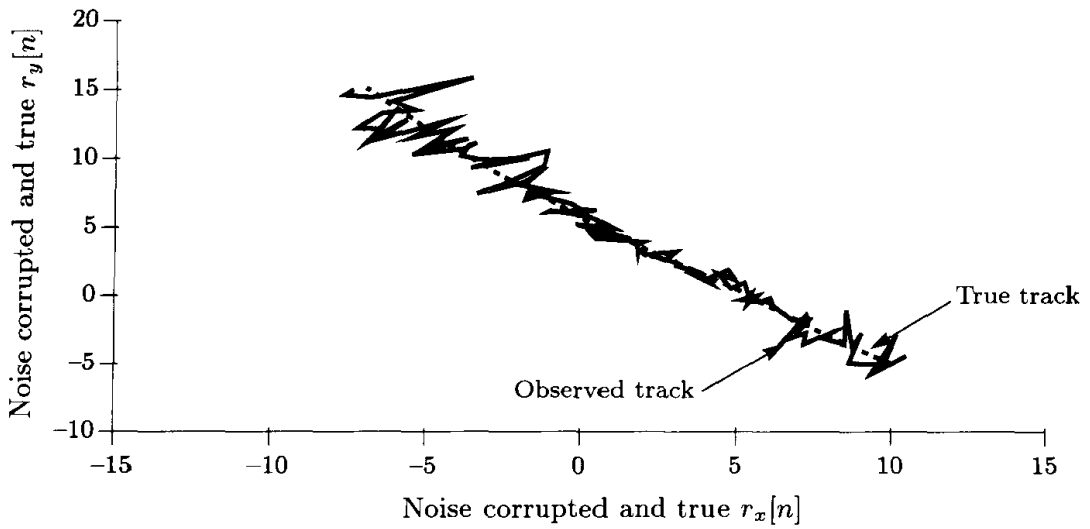


Figure 13.24 True and observed vehicle tracks

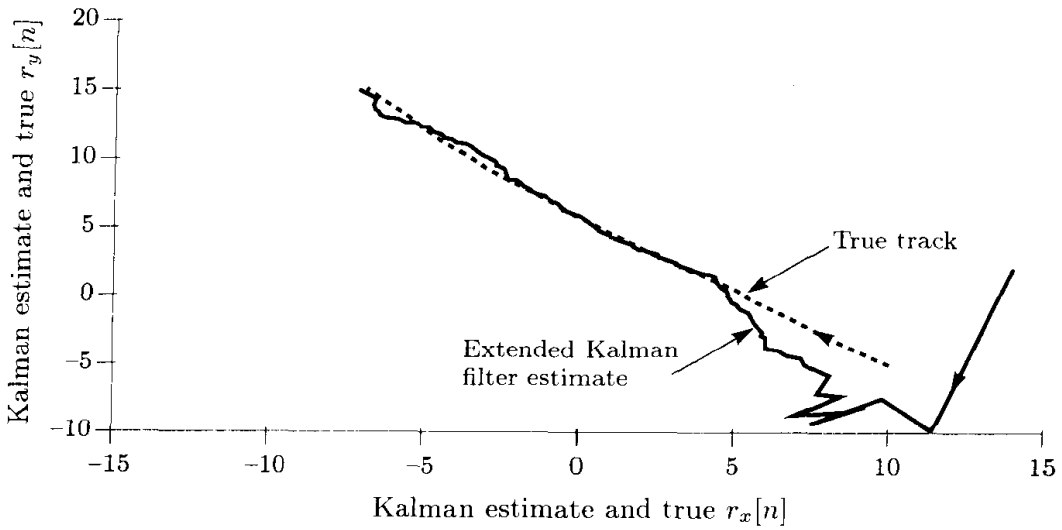


Figure 13.25 True and extended Kalman filter estimate

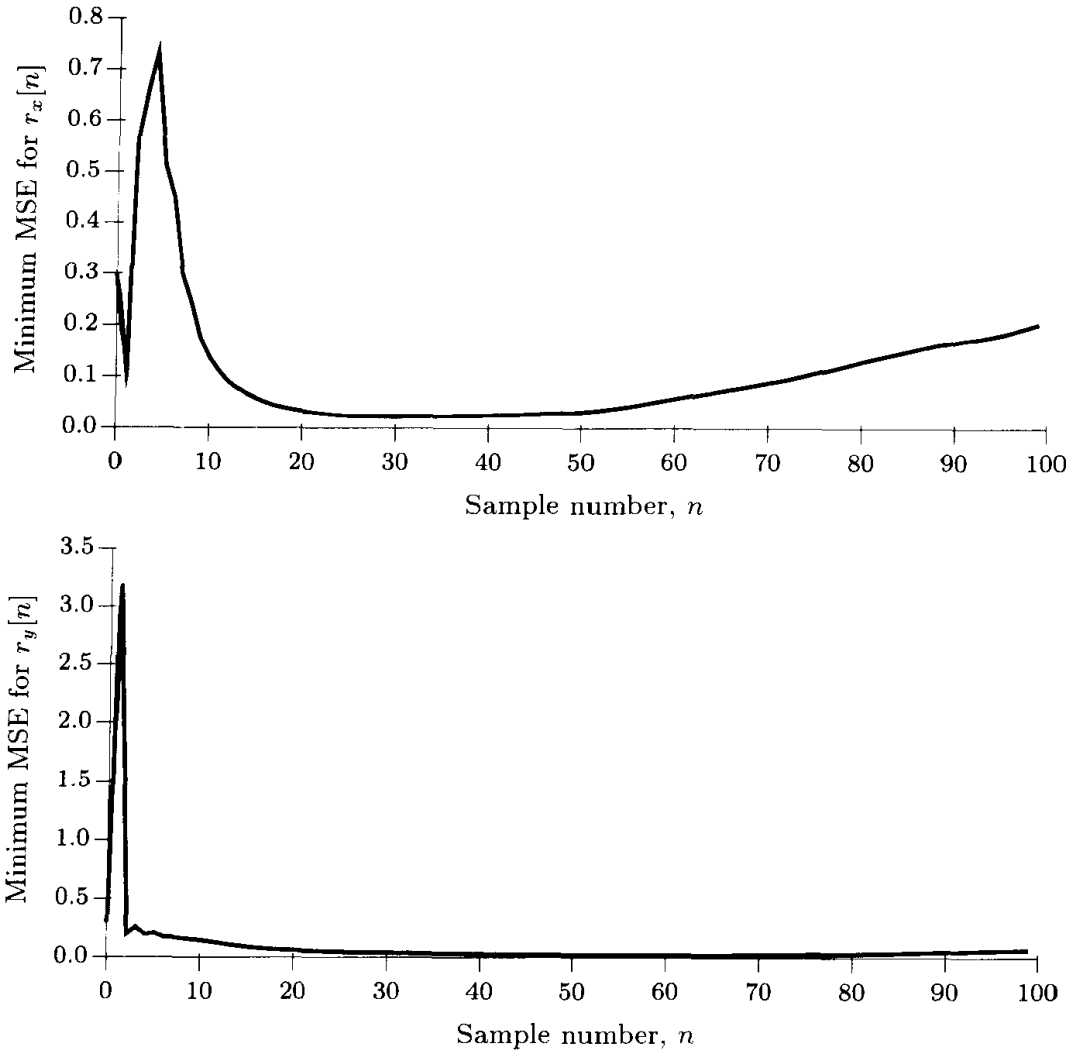


Figure 13.26 "Minimum" MSEs for $r_x[n]$ and $r_y[n]$

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Problems

13.1 A random process is Gaussian if for arbitrary samples $\{s[n_1], s[n_2], \ldots, s[n_k]\}$ and for any k the random vector $\mathbf{s} = [s[n_1]\, s[n_2] \ldots s[n_k]]^T$ is distributed according to a multivariate Gaussian PDF. If $s[n]$ is given by (13.2), prove that it is a Gaussian random process.

13.2 Consider a scalar Gauss-Markov process. Show that if $\mu_s = 0$ and $\sigma_s^2 = \sigma_u^2/(1 - a^2)$, then the process will be WSS for $n \geq 0$ and explain why this is so.

13.3 Plot the mean, variance, and steady-state covariance of a scalar Gauss-Markov process if $a = 0.98$, $\sigma_u^2 = 0.1$, $\mu_s = 5$, and $\sigma_s^2 = 1$. What is the PSD of the steady-state process?

13.4 For a scalar Gauss-Markov process derive a covariance propagation equation, i.e., a formula relating $c_s[m, n]$ to $c_s[n, n]$ for $m \geq n$. To do so first show that

$$c_s[m, n] = a^{m-n}\text{var}(s[n])$$

for $m \geq n$.

13.5 Verify (13.17) and (13.18) for the covariance matrix propagation of a vector Gauss-Markov process.

13.6 Show that the mean of a vector Gauss-Markov process in general will grow with n if any eigenvalue of $\mathbf{A}$ is greater than 1 in magnitude. What happens to the steady-state mean if all the eigenvalues are less than 1 in magnitude? To simplify matters assume that $\mathbf{A}$ is symmetric, so that it can be written as $\mathbf{A} = \sum_{i=1}^{p} \lambda_i \mathbf{v}_i \mathbf{v}_i^T$, where $\mathbf{v}_i$ is the ith eigenvector of $\mathbf{A}$ and λ_i is the corresponding *real* eigenvalue.

13.7 Show that

$$\mathbf{A}^n \mathbf{C}_s \mathbf{A}^{n^T} \rightarrow \mathbf{0}$$

as $n \rightarrow \infty$ if all the eigenvalues of $\mathbf{A}$ are less than 1 in magnitude. As in Problem 13.6, assume that $\mathbf{A}$ is symmetric. Hint: Examine $\mathbf{e}_i^T \mathbf{A}^n \mathbf{C}_s \mathbf{A}^{n^T} \mathbf{e}_j = [\mathbf{A}^n \mathbf{C}_s \mathbf{A}^{n^T}]_{ij}$, where $\mathbf{e}_i$ is a vector of all zeros except for a one in the ith element.

13.8 Consider the recursive difference equation (for $q < p$)

$$r[n] = -\sum_{k=1}^{p} a[k]r[n-k] + u[n] + \sum_{k=1}^{q} b[k]u[n-k] \qquad n \geq 0$$

where $u[n]$ is WGN with variance σ_u^2. (In steady-state this would be an ARMA process.) Let the state vector be defined as

$$\mathbf{s}[n-1] = \begin{bmatrix} r[n-p] \\ r[n-p+1] \\ \vdots \\ r[n-1] \end{bmatrix}$$

where $\mathbf{s}[-1] \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$ and is independent of $u[n]$. Also, define the vector driving noise sequence as

$$\mathbf{u}[n] = \begin{bmatrix} u[n-q] \\ u[n-q+1] \\ \vdots \\ u[n] \end{bmatrix}.$$

Rewrite the process as in (13.11). Explain why this is not a vector Gauss-Markov model by examining the assumptions on $\mathbf{u}[n]$.

13.9 For Problem 13.8 show that the process may alternatively be expressed by the difference equations

$$\begin{aligned} s[n] &= -\sum_{k=1}^{p} a[k]s[n-k] + u[n] \\ r[n] &= s[n] + \sum_{k=1}^{q} b[k]s[n-k] \end{aligned}$$

for $n \geq 0$. Assume that $\mathbf{s}[-1] = [s[-p]\, s[-p+1] \ldots s[-1]]^T \sim \mathcal{N}(\boldsymbol{\mu}_s, \mathbf{C}_s)$ and that we observe $x[n] = r[n] + w[n]$, where $w[n]$ is WGN with variance σ_n^2 , and $\mathbf{s}[-1], u[n]$, and $w[n]$ are independent. Show how to set up the vector state–scalar observation Kalman filter.

13.10 Assume that we observe $x[n] = A + w[n]$ for $n = 0, 1, \ldots$, where A is the realization of a random variable with PDF $\mathcal{N}(0, \sigma_A^2)$ and $w[n]$ is WGN with variance σ^2. Using the scalar state–scalar observation Kalman filter find a sequential estimator of A based on $\{x[0], x[1], \ldots, x[n]\}$ or $\hat{A}[n]$. Solve explicitly for $\hat{A}[n]$, the Kalman gain, and the minimum MSE.

13.11 In this problem we implement a scalar state–scalar observation Kalman filter (see (13.38)-(13.42)). A computer solution is advised. If $a = 0.9$, $\sigma_u^2 = 1$, $\mu_s = 0$, $\sigma_s^2 = 1$, find the Kalman gain and minimum MSE if

a. $\sigma_n^2 = (0.9)^n$

b. $\sigma_n^2 = 1$

c. $\sigma_n^2 = (1.1)^n$.

Explain your results. Using a Monte Carlo computer simulation generate a realization of the signal and noise and apply your Kalman filter to estimation of the signal for all three cases. Plot the signal as well as the Kalman filter estimate.

13.12 For the scalar state–scalar observation Kalman filter assume that $\sigma_n^2 = 0$ for all n so that we observe $s[n]$ directly. Find the innovation sequence. Is it white?

13.13 In this problem we show that the same set of equations result for the scalar state–scalar observation Kalman filter even if $E(s[-1]) \neq 0$. To do so let $s'[n] = s[n] - E(s[n])$ and $x'[n] = x[n] - E(x[n])$, so that equations (13.38)–(13.42) apply for $s'[n]$. Now determine the corresponding equations for $s[n]$. Recall that the MMSE estimator of $\theta + c$ for c a constant is $\hat{\theta} + c$, where $\hat{\theta}$ is the MMSE of θ.

13.14 Prove that for the scalar state–scalar observation Kalman filter

$$M[n|n-1] > M[n-1|n-1]$$

for large enough n or in steady-state. Why is this reasonable?

13.15 Prove that the optimal l-step predictor for a scalar Gauss-Markov process $s[n]$ is

$$\hat{s}[n+l|n] = a^l \hat{s}[n|n]$$

where $\hat{s}[n|n]$ and $\hat{s}[n+l|n]$ are based on $\{x[0], x[1], \ldots, x[n]\}$.

13.16 Find the transfer function of the steady-state, scalar state–scalar observation Kalman filter if $a = 0.8$, $\sigma_u^2 = 1$, and $\sigma^2 = 1$. Give its time domain form as a recursive difference equation.

13.17 For the scalar state–scalar observation Kalman filter let $a = 0.9$, $\sigma_u^2 = 1$, $\sigma^2 = 1$ and find the steady-state gain and minimum MSE by running the filter until convergence, i.e., compute equations (13.39), (13.40), and (13.42). Compare your results to those given in Section 13.5.

13.18 Assume we observe the data

$$x[k] = Ar^k + w[k]$$

for $k = 0, 1, \ldots, n$, where A is the realization of a random variable with PDF $\mathcal{N}(\mu_A, \sigma_A^2)$, $0 < r < 1$, and the $w[k]$'s are samples of WGN with variance σ^2. Furthermore, assume that A is independent of the $w[k]$'s. Find the sequential MMSE estimator of A based on $\{x[0], x[1], \ldots, x[n]\}$.

13.19 Consider the vector state–vector observation Kalman filter for which $\mathbf{H}[n]$ is assumed to be invertible. If a particular observation is noiseless, so that $\mathbf{C}[n] = \mathbf{0}$, find $\hat{\mathbf{s}}[n|n]$ and explain your results. What happens if $\mathbf{C}[n] \to \infty$?

13.20 Prove that the optimal l-step predictor for a vector Gauss-Markov process $\mathbf{s}[n]$ is

$$\hat{\mathbf{s}}[n+l|n] = \mathbf{A}^l \hat{\mathbf{s}}[n|n]$$

where $\hat{\mathbf{s}}[n|n]$ and $\hat{\mathbf{s}}[n+l|n]$ are based on $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n]\}$. Use the vector state–vector observation Kalman filter.

13.21 In this problem we set up an extended Kalman filter for the frequency tracking application. Specifically, we wish to track the frequency of a sinusoid in noise. The frequency is assumed to follow the model

$$f_0[n] = af_0[n-1] + u[n] \qquad n \geq 0$$

where $u[n]$ is WGN with variance σ_u^2 and $f_0[-1]$ is distributed according to $\mathcal{N}(\mu_{f_0}, \sigma_{f_0}^2)$ and is independent of $u[n]$. The observed data are

$$x[n] = \cos(2\pi f_0[n]) + w[n] \qquad n \geq 0$$

where $w[n]$ is WGN with variance σ^2 and is independent of $u[n]$ and $f_0[-1]$. Write down the extended Kalman filter equations for this problem.

13.22 For the vehicle position model in Example 13.4 we had

$$v_x[n] = v_x[n-1] + u_x[n]$$

where $u_x[n]$ is WGN with variance σ_u^2. Find the variance of $v_x[n]$ to show that it increases with n. What might be a more suitable model for $v_x[n]$? (See also [Anderson and Moore 1979] for a further discussion of the modeling issue.)

13.23 For the scalar state–scalar observation Kalman filter find an expression relating $M[n|n]$ to $M[n-1|n-1]$. Now, let $a = 0.9$, $\sigma_u^2 = 1$, and $\sigma_n^2 = n+1$. If $M[-1|-1] = 1$, determine $M[n|n]$ for $n \geq 0$. Explain your results.

Appendix 13A

Vector Kalman Filter Derivation

We derive the vector state–vector observation Kalman filter with the vector state–scalar observation being a special case. Theorem 13.2 contains the modeling assumptions. In our derivation we will assume $\boldsymbol{\mu}_s = \mathbf{0}$, so that all random variables are zero mean. With this assumption the vector space viewpoint is applicable. For $\boldsymbol{\mu}_s \neq \mathbf{0}$ it can be shown that the same equations result. The reason for this is a straightforward generalization of the comments made at the end of Appendix 12A (see also Problem 13.13).

The properties of MMSE estimators that we will use are

1. The MMSE estimator of $\boldsymbol{\theta}$ based on two uncorrelated data samples $\mathbf{x}_1$ and $\mathbf{x}_2$, assuming jointly Gaussian statistics (see Section 11.4), is

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}|\mathbf{x}_1, \mathbf{x}_2) \\ &= E(\boldsymbol{\theta}|\mathbf{x}_1) + E(\boldsymbol{\theta}|\mathbf{x}_2) \end{aligned}$$

if $\boldsymbol{\theta}, \mathbf{x}_1, \mathbf{x}_2$ are zero mean.

2. The MMSE estimator is linear in that if $\boldsymbol{\theta} = \mathbf{A}_1\boldsymbol{\theta}_1 + \mathbf{A}_2\boldsymbol{\theta}_2$, then

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}|\mathbf{x}) \\ &= E(\mathbf{A}_1\boldsymbol{\theta}_1 + \mathbf{A}_2\boldsymbol{\theta}_2|\mathbf{x}) \\ &= \mathbf{A}_1 E(\boldsymbol{\theta}_1|\mathbf{x}) + \mathbf{A}_2 E(\boldsymbol{\theta}_2|\mathbf{x}) \\ &= \mathbf{A}_1\hat{\boldsymbol{\theta}}_1 + \mathbf{A}_2\hat{\boldsymbol{\theta}}_2. \end{aligned}$$

The derivation follows that for the scalar state–scalar observation Kalman filter with obvious adjustments for vector quantities. We assume $\boldsymbol{\mu}_s = \mathbf{0}$ so that all random vectors are zero mean. The MMSE estimator of $\mathbf{s}[n]$ based on $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n]\}$ is the mean of the posterior PDF

$$\hat{\mathbf{s}}[n|n] = E(\mathbf{s}[n]|\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n]). \tag{13A.1}$$

But $\mathbf{s}[n], \mathbf{x}[0], \ldots, \mathbf{x}[n]$ are jointly Gaussian since from (13.13) $\mathbf{s}[n]$ depends linearly on $\{\mathbf{s}[-1], \mathbf{u}[0], \ldots, \mathbf{u}[n]\}$ and $\mathbf{x}[n]$ depends linearly on $\mathbf{s}[n], \mathbf{w}[n]$. Hence, we have a linear

dependence on the set of random vectors $S = \{\mathbf{s}[-1], \mathbf{u}[0], \ldots, \mathbf{u}[n], \mathbf{w}[0], \ldots, \mathbf{w}[n]\}$, where each random vector is independent of the others. As a result, the vectors in S are jointly Gaussian and any linear transformation also produces jointly Gaussian random vectors. Now, from (10.24) with zero means we have

$$\hat{\mathbf{s}}[n|n] = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \tag{13A.2}$$

where $\boldsymbol{\theta} = \mathbf{s}[n]$ and $\mathbf{x} = [\mathbf{x}^T[0]\,\mathbf{x}^T[1]\ldots\mathbf{x}^T[n]]^T$ and is seen to be linear in $\mathbf{x}$. To determine a recursive in time algorithm we appeal to the vector space approach. Let

$$\mathbf{X}[n] = \begin{bmatrix} \mathbf{x}^T[0] & \mathbf{x}^T[1] & \ldots & \mathbf{x}^T[n] \end{bmatrix}$$

and let $\tilde{\mathbf{x}}[n] = \mathbf{x}[n] - \hat{\mathbf{x}}[n|n-1]$ be the innovation or the error incurred by linearly estimating $\mathbf{x}[n]$ based on $\mathbf{X}[n-1]$. Now we can replace (13A.1) by

$$\hat{\mathbf{s}}[n|n] = E(\mathbf{s}[n]|\mathbf{X}[n-1], \tilde{\mathbf{x}}[n])$$

since $\mathbf{x}[n]$ is recoverable from $\mathbf{X}[n-1]$ and $\tilde{\mathbf{x}}[n]$ (see comments in Section 13.4). Because $\mathbf{X}[n-1]$ and $\tilde{\mathbf{x}}[n]$ are uncorrelated, we have from property 1

$$\begin{aligned} \hat{\mathbf{s}}[n|n] &= E(\mathbf{s}[n]|\mathbf{X}[n-1]) + E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]) \\ &= \hat{\mathbf{s}}[n|n-1] + E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]). \end{aligned}$$

The predicted sample is found from (13.12) as

$$\begin{aligned} \hat{\mathbf{s}}[n|n-1] &= E(\mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n]|\mathbf{X}[n-1]) \\ &= \mathbf{A}\hat{\mathbf{s}}[n-1|n-1] + \mathbf{B}E(\mathbf{u}[n]|\mathbf{X}[n-1]) \\ &= \mathbf{A}\hat{\mathbf{s}}[n-1|n-1] \end{aligned}$$

since the second term is zero. To see why, note that

$$E(\mathbf{u}[n]|\mathbf{X}[n-1]) = E(\mathbf{u}[n]) = \mathbf{0}$$

since from (13.13) $\mathbf{X}[n-1]$ depends on $\{\mathbf{s}[-1], \mathbf{u}[0], \ldots, \mathbf{u}[n-1], \mathbf{w}[0], \mathbf{w}[1], \ldots, \mathbf{w}[n-1]\}$ which are all independent of $\mathbf{u}[n]$. We now have

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]) \tag{13A.3}$$

where

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{A}\hat{\mathbf{s}}[n-1|n-1]. \tag{13A.4}$$

To determine $E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n])$ we recall that it is the MMSE estimator of $\mathbf{s}[n]$ based on $\tilde{\mathbf{x}}[n]$. Since $\mathbf{s}[n]$ and $\tilde{\mathbf{x}}[n]$ are jointly Gaussian, we have from (10.24)

$$E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]) = \mathbf{C}_{s\tilde{x}}\mathbf{C}_{\tilde{x}\tilde{x}}^{-1}\tilde{\mathbf{x}}[n].$$

Letting $\mathbf{K}[n] = \mathbf{C}_{s\tilde{x}}\mathbf{C}_{\tilde{x}\tilde{x}}^{-1}$ be the gain matrix, this is rewritten as

$$E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]) = \mathbf{K}[n](\mathbf{x}[n] - \hat{\mathbf{x}}[n|n-1]).$$

But

$$\mathbf{x}[n] = \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n]$$

so that from property 2

$$\begin{aligned} \hat{\mathbf{x}}[n|n-1] &= \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1] + \hat{\mathbf{w}}[n|n-1] \\ &= \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1] \end{aligned} \tag{13A.5}$$

since $\hat{\mathbf{w}}[n|n-1] = \mathbf{0}$ due to the independence of $\mathbf{w}[n]$ with $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[n-1]\}$. Thus,

$$E(\mathbf{s}[n]|\tilde{\mathbf{x}}[n]) = \mathbf{K}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])$$

and from (13A.3) we now have

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]) \tag{13A.6}$$

where

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{A}\hat{\mathbf{s}}[n-1|n-1].$$

The gain matrix $\mathbf{K}[n]$ is

$$\begin{aligned} \mathbf{K}[n] &= \mathbf{C}_{s\tilde{x}}\mathbf{C}_{\tilde{x}\tilde{x}}^{-1} \\ &= E(\mathbf{s}[n]\tilde{\mathbf{x}}^T[n])E(\tilde{\mathbf{x}}[n]\tilde{\mathbf{x}}^T[n])^{-1}. \end{aligned}$$

To evaluate it we will need the results

1.

$$E\left[\mathbf{s}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right] =$$

$$E\left[(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1])(\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right].$$

2.

$$E\left[\mathbf{w}[n](\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1])^T\right] = \mathbf{0}.$$

The first result follows from the innovation $\tilde{\mathbf{x}}[n] = \mathbf{x}[n] - \hat{\mathbf{x}}[n|n-1] = \mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]$ being uncorrelated with the past data and hence with $\hat{\mathbf{s}}[n|n-1]$. The second result is due to the assumption that $\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1]$ is a linear combination of $\{\mathbf{s}[-1], \mathbf{u}[0], \ldots, \mathbf{u}[n], \mathbf{w}[0], \ldots, \mathbf{w}[n-1]\}$, which is independent of $\mathbf{w}[n]$. We first examine $\mathbf{C}_{s\tilde{x}}$ using results 1 and 2 and also (13A.5).

$$\begin{aligned} \mathbf{C}_{s\tilde{x}} &= E\left[\mathbf{s}[n](\mathbf{x}[n] - \hat{\mathbf{x}}[n|n-1])^T\right] \\ &= E\left[\mathbf{s}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right] \\ &= E\left[(\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1])(\mathbf{H}[n]\mathbf{s}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1] + \mathbf{w}[n])^T\right] \\ &= \mathbf{M}[n|n-1]\mathbf{H}^T[n] \end{aligned}$$

and using result 2 and (13A.5) again

$$\begin{aligned}
\mathbf{C}_{\tilde{x}\tilde{x}} &= E\left[(\mathbf{x}[n]-\hat{\mathbf{x}}[n|n-1])(\mathbf{x}[n]-\hat{\mathbf{x}}[n|n-1])^T\right] \\
&= E\left[(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right] \\
&= E\left[(\mathbf{H}[n]\mathbf{s}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]+\mathbf{w}[n])\right. \\
&\quad \left.\cdot\,(\mathbf{H}[n]\mathbf{s}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]+\mathbf{w}[n])^T\right] \\
&= \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n]+\mathbf{C}[n].
\end{aligned}$$

Thus, the Kalman gain matrix is

$$\mathbf{K}[n]=\mathbf{M}[n|n-1]\mathbf{H}^T[n](\mathbf{C}[n]+\mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n])^{-1}.$$

To evaluate this we require $\mathbf{M}[n|n-1]$, so that by using (13A.4)

$$\begin{aligned}
\mathbf{M}[n|n-1] &= E\left[(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])^T\right] \\
&= E\left[(\mathbf{A}\mathbf{s}[n-1]+\mathbf{B}\mathbf{u}[n]-\mathbf{A}\hat{\mathbf{s}}[n-1|n-1])\right. \\
&\quad \left.\cdot\,(\mathbf{A}\mathbf{s}[n-1]+\mathbf{B}\mathbf{u}[n]-\mathbf{A}\hat{\mathbf{s}}[n-1|n-1])^T\right] \\
&= E\left[(\mathbf{A}(\mathbf{s}[n-1]-\hat{\mathbf{s}}[n-1|n-1])+\mathbf{B}\mathbf{u}[n])\right. \\
&\quad \left.\cdot\,(\mathbf{A}(\mathbf{s}[n-1]-\hat{\mathbf{s}}[n-1\mid n-1])+\mathbf{B}\mathbf{u}[n])^T\right]
\end{aligned}$$

and using the result

$$E\left[(\mathbf{s}[n-1]-\hat{\mathbf{s}}[n-1|n-1])\mathbf{u}^T[n]\right]=\mathbf{0} \tag{13A.7}$$

we have

$$\mathbf{M}[n|n-1]=\mathbf{A}\mathbf{M}[n-1|n-1]\mathbf{A}^T+\mathbf{B}\mathbf{Q}\mathbf{B}^T.$$

The equation (13A.7) is true because $\mathbf{s}[n-1]-\hat{\mathbf{s}}[n-1|n-1]$ depends on $\{\mathbf{s}[-1],\mathbf{u}[0],\ldots,\mathbf{u}[n-1],\mathbf{w}[0],\ldots,\mathbf{w}[n-1]\}$, which are independent of $\mathbf{u}[n]$. Finally, to determine the recursion for $\mathbf{M}[n|n]$ we have from (13A.6)

$$\begin{aligned}
\mathbf{M}[n|n] &= E\left[(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n])(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n])^T\right] \\
&= E\left[(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1]-\mathbf{K}[n](\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]))\right. \\
&\quad \left.\cdot\,(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1]-\mathbf{K}[n](\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]))^T\right] \\
&= E\left[(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])^T\right] \\
&\quad -E\left[(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right]\mathbf{K}^T[n] \\
&\quad -\mathbf{K}[n]E\left[(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])(\mathbf{s}[n]-\hat{\mathbf{s}}[n|n-1])^T\right] \\
&\quad +\mathbf{K}[n]E\left[(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])(\mathbf{x}[n]-\mathbf{H}[n]\hat{\mathbf{s}}[n|n-1])^T\right]\mathbf{K}^T[n].
\end{aligned}$$

The first term is $\mathbf{M}[n|n-1]$, the second expectation is $\mathbf{C}_{s\tilde{x}}$ using result 1, the third expectation is $\mathbf{C}_{s\tilde{x}}^T$, and the last expectation is $\mathbf{C}_{\tilde{x}\tilde{x}}$. From the derivation for $\mathbf{K}[n]$ we have that

$$\mathbf{K}[n]=\mathbf{C}_{s\tilde{x}}\mathbf{C}_{\tilde{x}\tilde{x}}^{-1}$$

where

$$\mathbf{C}_{s\tilde{x}} = \mathbf{M}[n|n-1]\mathbf{H}^T[n].$$

Hence,

$$\begin{aligned}
\mathbf{M}[n|n] &= \mathbf{M}[n|n-1] - \mathbf{C}_{s\tilde{x}}\mathbf{K}^T[n] - \mathbf{K}[n]\mathbf{C}_{s\tilde{x}}^T + \mathbf{K}[n]\mathbf{C}_{\tilde{x}\tilde{x}}\mathbf{K}^T[n] \\
&= \mathbf{M}[n|n-1] - \mathbf{C}_{s\tilde{x}}\mathbf{K}^T[n] - \mathbf{K}[n]\mathbf{C}_{s\tilde{x}}^T + \mathbf{C}_{s\tilde{x}}\mathbf{C}_{\tilde{x}\tilde{x}}^{-1}\mathbf{C}_{s\tilde{x}}^T \\
&= \mathbf{M}[n|n-1] - \mathbf{K}[n]\mathbf{C}_{s\tilde{x}}^T \\
&= (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1]
\end{aligned}$$

which completes the derivation.

Appendix 13B

Extended Kalman Filter Derivation

To derive the extended Kalman filter equations we need to first determine the equations for a modified state model that has a known deterministic input or

$$\mathbf{s}[n] = \mathbf{A}\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] + \mathbf{v}[n] \qquad n \geq 0 \tag{13B.1}$$

where $\mathbf{v}[n]$ is known. The presence of $\mathbf{v}[n]$ at the input will produce a deterministic component in the output, so that the mean of $\mathbf{s}[n]$ will no longer be zero. We assume that $E(\mathbf{s}[-1]) = \mathbf{0}$, so that if $\mathbf{v}[n] = \mathbf{0}$, then $E(\mathbf{s}[n]) = \mathbf{0}$. Hence, the effect of the deterministic input is to produce a nonzero mean signal vector

$$\mathbf{s}[n] = \mathbf{s}'[n] + E(\mathbf{s}[n]) \tag{13B.2}$$

where $\mathbf{s}'[n]$ is the value of $\mathbf{s}[n]$ when $\mathbf{v}[n] = \mathbf{0}$. Then, we can write a state equation for the zero mean signal vector $\mathbf{s}'[n] = \mathbf{s}[n] - E(\mathbf{s}[n])$ as

$$\mathbf{s}'[n] = \mathbf{A}\mathbf{s}'[n-1] + \mathbf{B}\mathbf{u}[n].$$

Note that the mean satisfies

$$E(\mathbf{s}[n]) = \mathbf{A}E(\mathbf{s}[n-1]) + \mathbf{v}[n] \tag{13B.3}$$

which follows from (13B.1). Likewise, the observation equation is

$$\begin{aligned} \mathbf{x}[n] &= \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] \\ &= \mathbf{H}[n]\mathbf{s}'[n] + \mathbf{w}[n] + \mathbf{H}[n]E(\mathbf{s}[n]) \end{aligned}$$

or letting

$$\mathbf{x}'[n] = \mathbf{x}[n] - \mathbf{H}[n]E(\mathbf{s}[n]),$$

we have the usual observation equation

$$\mathbf{x}'[n] = \mathbf{H}[n]\mathbf{s}'[n] + \mathbf{w}[n].$$

The Kalman filter for $\mathbf{s}'[n]$ can be found using (13.58)–(13.62) with $\mathbf{x}[n]$ replaced by $\mathbf{x}'[n]$. Then, the MMSE estimator of $\mathbf{s}[n]$ can easily be found by the relations

$$\begin{aligned}\hat{\mathbf{s}}'[n|n-1] &= \hat{\mathbf{s}}[n|n-1] - E(\mathbf{s}[n]) \\ \hat{\mathbf{s}}'[n-1|n-1] &= \hat{\mathbf{s}}[n-1|n-1] - E(\mathbf{s}[n-1]).\end{aligned}$$

Using these in (13.58), we have for the prediction equation

$$\hat{\mathbf{s}}'[n|n-1] = \mathbf{A}\hat{\mathbf{s}}'[n-1|n-1]$$

or

$$\hat{\mathbf{s}}[n|n-1] - E(\mathbf{s}[n]) = \mathbf{A}\left(\hat{\mathbf{s}}[n-1|n-1] - E(\mathbf{s}[n-1])\right)$$

and from (13B.3) this reduces to

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{A}\hat{\mathbf{s}}[n-1|n-1] + \mathbf{v}[n].$$

For the correction we have from (13.61)

$$\hat{\mathbf{s}}'[n|n] = \hat{\mathbf{s}}'[n|n-1] + \mathbf{K}[n](\mathbf{x}'[n] - \mathbf{H}[n]\hat{\mathbf{s}}'[n|n-1])$$

or

$$\begin{aligned}&\hat{\mathbf{s}}[n|n] - E(\mathbf{s}[n]) \\ &= \hat{\mathbf{s}}[n|n-1] - E(\mathbf{s}[n]) + \mathbf{K}[n]\left[\mathbf{x}[n] - \mathbf{H}[n]E(\mathbf{s}[n]) - \mathbf{H}[n](\hat{\mathbf{s}}[n|n-1] - E(\mathbf{s}[n]))\right].\end{aligned}$$

This reduces to the usual equation

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]).$$

The Kalman gain as well as the MSE matrices remain the same since the MMSE estimator will not incur any additional error due to known constants. Hence, the only revision is in the prediction equation.

Returning to the extended Kalman filter, we also have the modified observation equation

$$\mathbf{x}[n] = \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] + \mathbf{z}[n]$$

where $\mathbf{z}[n]$ is known. With $\mathbf{x}'[n] = \mathbf{x}[n] - \mathbf{z}[n]$ we have the usual observation equation. Hence, our two equations become, upon replacing $\mathbf{A}$ by $\mathbf{A}[n]$,

$$\begin{aligned}\hat{\mathbf{s}}[n|n-1] &= \mathbf{A}[n-1]\hat{\mathbf{s}}[n-1|n-1] + \mathbf{v}[n] \\ \hat{\mathbf{s}}[n|n] &= \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{z}[n] - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]).\end{aligned}$$

But

$$\begin{aligned}\mathbf{v}[n] &= \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) - \mathbf{A}[n-1]\hat{\mathbf{s}}[n-1|n-1] \\ \mathbf{z}[n] &= \mathbf{h}(\hat{\mathbf{s}}[n|n-1]) - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]\end{aligned}$$

so that finally

$$\begin{aligned}\hat{\mathbf{s}}[n|n-1] &= \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) \\ \hat{\mathbf{s}}[n|n] &= \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{\mathbf{s}}[n|n-1])).\end{aligned}$$